Mont Terri CD-A Niche 4 OGS Model Notes

Mathematical TRM Model, OGS Realization, and Measurement Links

OGS modelling note

June 6, 2026

Abstract

This note records the CD-A Niche 4 model in two layers. The first layer is the mathematical Thermo–Richards–Mechanics model: physical setting, unknowns, constitutive assumptions, governing equations, boundary data, and initial state. The second layer is the OGS realization of the same model: project-file blocks, mesh fields, XML parameters, output arrays, and observation-facing quantities. This separation is deliberate. It lets us discuss the model as a scientific object before we decide which material fields, boundary curves, and measurement operators are safe to release for inversion or validation.

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Chapter 1

Mathematical TRM Model and OGS Realization

1.1 Purpose and Evidence Boundary

We use a two-layer description of the CD-A Niche 4 reference model. The first layer is the mathematical Thermo–Richards–Mechanics (TRM) model: domain, unknowns, constitutive laws, boundary data, and initial state. The second layer is the OGS realization of that model through project-file blocks, XML parameters, mesh fields, curves, time stepping, and VTU outputs. This order is deliberate. It fixes the scientific model before we discuss how each field is passed into OGS or read from OGS results.

The file-level evidence for the implementation is the transferred OGS project `cd_a_open_niche_quad.prj`. Its included process, process-variable, parameter, media, time-loop, solver, and curve XML files are model inputs, not literature sources; when a statement is model-specific, the relevant XML tag or path is stated in the text or tables. For the equations we use the OGS TRM formulation without vapor diffusion, the coupled THM notation of Wang et al. [24, Sec. 2.1–2.3, Eqs. 1, 7–9], and the OGS documentation [19, "Governing equations without vapor diffusion"]. If the OGS project files contain an inactive parameter, commented block, or implementation mismatch, we record it as a gate. We do not promote such entries into the active physics.

1.2 Physical Setting and Modelled Phenomena

The CD-A experiment studies the coupled response of Opalinus Clay around twin niches in the Mont Terri Rock Laboratory. The relevant rock is a bedded claystone with strong hydro-mechanical coupling, structural controls, and a near-niche excavation damaged zone (EDZ). The CD-A work focuses on humidity-driven water-content, pressure, and deformation evolution in the formation [27, abstract] [29, abstract, Sec. 2, Table 1] [30, abstract and Secs. 1–2]. Broader Mont Terri studies show that Opalinus Clay stiffness, strength, pore-pressure response, and damage-zone development depend on suction, bedding, in-situ stress, and fault or fracture structure [1, abstract, Secs. 2–3.3]. EDZ studies further show that hydraulic conductivity and fracture aperture can differ by orders of magnitude from the intact claystone and can evolve through swelling, closure, and self-sealing [11, abstract and Sec. 3.1.1] [14, abstract, Secs. 2.1–2.5].

The reference OGS model represents a controlled subset of this physics. It includes Richards pressure diffusion, capillary retention, relative permeability, porosity storage, orthotropic linear elasticity, Biot/Bishop pore-stress coupling, saturation-dependent swelling stress, and heat-balance terms. It does not explicitly represent vapor diffusion, discrete fracture flow, chemical alteration, sealing/healing kinetics, staged excavation, or full three-dimensional fault geometry. These omissions matter near the niche and in the EDZ, but they remain external gates or later scenario fields in this chapter.

1.3 Mathematical Model

1.3.1 Domain, Unknowns, and State Fields

Let $\Omega \subset \mathbb{R}^2$ be the current two-dimensional CD-A Niche 4 slice, and let $I = [0, t_{\text{end}}]$ be the simulated time interval. The primary unknowns are displacement $\boldsymbol{u} : \Omega \times I \to \mathbb{R}^2$, liquid pressure $p_l : \Omega \times I \to \mathbb{R}$, and temperature $T : \Omega \times I \to \mathbb{R}$. Liquid saturation, Darcy velocity, stress, strain, swelling stress, and water-content proxies are state-dependent or post-processed quantities. Tables 1.1–1.3 separate the notation into unknown and output fields, material input fields, and intermediate or derived quantities before they enter the equations.

Table 1.1: Unknowns and OGS output fields used in the reference model.

Field	Symbol	Unit	Mathematical meaning and OGS status
Displacement	$\boldsymbol{u} = (u_x, u_y)$	m	Primary mechanical unknown and current OGS output field. It describes deformation of the continuum slice relative to the chosen numerical initial state.
Liquid pressure	p_l	Pa	Primary hydraulic unknown and current OGS output field. In the Richards convention used here, negative liquid pressure means positive suction relative to the gas/reference pressure.
Temperature	T	K	Primary thermal unknown in the full TRM process and current OGS output field. In the active parametrization it is constrained to $T_0 = 298.15\,\mathrm{K}$ throughout the domain and time interval.
Liquid saturation	S_l	–	Secondary OGS state/output field computed from pressure through the retention law. It is the mobile OGS liquid occupancy of the pore volume, not total hydrogen-bearing water in the clay.

Table 1.2: Material fields and constitutive inputs to the active model.

Input field or parameter	Symbol	Unit	Material meaning and OGS role
Intrinsic permeability	$\boldsymbol{K}(\boldsymbol{x})$	m^2	Material tensor controlling Darcy mobility before multiplication by relative permeability and division by viscosity. The current workflow supplies it through the current OGS mesh field `k_i_rd`.
Porosity	$n(\boldsymbol{x})$	–	Continuum pore-volume fraction used in storage, thermal mixing, and the model-side water-content proxy. It is supplied through `n_rd` and is fixed at 0.105 in the current field package.
Elasticity tensor	$\mathbb{C}$	Pa	Orthotropic linear-elastic stiffness used in the mechanical constitutive law. It is a fixed constitutive input, not a released field in the present workflow.
Biot coefficient and Bishop law	α_B, $b(S_l)$	–	Fixed pore-pressure coupling inputs. In the active XML, $\alpha_B = 1$ and $b(S_l) = S_l$.
Liquid properties	$\rho_l(T, p_l)$, $\rho_l^0(p_l)$, μ_l	$\mathrm{kg\,m^{-3}}$, $\mathrm{Pa\,s}$	Liquid density and viscosity used in storage and Darcy mobility. Under the active isothermal constraint, density depends on pressure but not on temperature.

Table 1.3: Intermediate and derived quantities used by equations and observation operators.

Quantity	Symbol	Unit	Mathematical meaning
Capillary pressure	$p_c = -p_l$	Pa	Retention-curve argument under the active sign convention.

Quantity	Symbol	Unit	Mathematical meaning
Effective saturation	$S_e(p_c)$	–	Normalized saturation variable used by the van Genuchten retention and relative-permeability laws.
Relative permeability	$k_{\mathrm{rel}}(S_l)$	–	Saturation-dependent mobility reduction in the liquid phase. It is computed from the fixed relative-permeability law, not fitted independently.
Darcy velocity and mass flux	$\boldsymbol{v}_l$, $\boldsymbol{q}_l = \rho_l \boldsymbol{v}_l$	$\mathrm{m\,s^{-1}}$, $\mathrm{kg\,m^{-2}\,s^{-1}}$	Derived hydraulic transport quantities. They are not pore velocity unless divided by an explicitly chosen pore-volume factor.
Strain and stress	$\boldsymbol{\varepsilon}(\boldsymbol{u})$, $\boldsymbol{\varepsilon}^{\mathrm{th}}(T)$, $\boldsymbol{\sigma}$	–, Pa	Mechanical fields derived from displacement, orthotropic elasticity, swelling strain, thermal strain, and Bishop/Biot pore-pressure coupling.
Swelling strain or stress contribution	$\boldsymbol{\varepsilon}^{\mathrm{sw}}$, p_{sw}	–, Pa	Saturation-dependent mechanical contribution used inside the active constitutive law.
Volumetric water-content proxy	$\theta = nS_l$	–	Model-derived mobile-water proxy. It needs an observation operator before it can be compared with NMR, ERT, or Taupe/TDR.

We assume a single liquid phase in a porous solid, Richards-type capillary pressure closure, no explicit gas-pressure unknown, and no active vapor diffusion. The reference body force is zero. Temperature is retained in the full TRM statement because it is an OGS primary variable and appears in density, heat storage, and thermo-mechanical coupling [19, "Governing equations without vapor diffusion" and "Process variables"]. Section 1.3.6 then records the reduced isothermal model implied by the temperature initial and boundary entries in `02_process_variables_TRM.xml` and `03_parameters_TRM.xml`.

1.3.2 Liquid Flow, Retention, and Relative Permeability

The liquid velocity is the Darcy velocity. Here k_{rel} is relative permeability, μ_l is liquid viscosity, $\boldsymbol{K}$ is intrinsic permeability, ρ_l is liquid density, and $\boldsymbol{q}_l$ is liquid mass flux. With the specific body force disabled, the reference equation is

$$\boldsymbol{v}_l = -\frac{k_{\mathrm{rel}}(S_l)}{\mu_l} \boldsymbol{K} \nabla p_l, \qquad \boldsymbol{q}_l = \rho_l \boldsymbol{v}_l. \tag{1.1}$$

The permeability tensor $\boldsymbol{K}$ is intrinsic permeability in m^2. It is not hydraulic conductivity. Hydraulic conductivity is a derived quantity after choosing liquid density, viscosity, relative permeability, and saturation state. The factor nS_l belongs to storage, thermal mixing, and the water-content proxy; it is not part of the Darcy velocity in (1.1).

The active pressure convention is

$$p_c := -p_l. \tag{1.2}$$

Negative liquid pressure therefore means positive suction/capillary pressure under the gas/reference convention used by the Richards formulation. For $p_c > 0$, the configured van Genuchten retention law is written with effective saturation S_e. The parameters S_{lr}, S_{gr}, m, and p_b are the residual liquid saturation, residual gas saturation, retention exponent, and pressure scale listed in Table 1.4:

$$S_e(p_c) = \left[1 + \left(\frac{p_c}{p_b}\right)^{1/(1-m)}\right]^{-m}, \qquad S_l = S_{lr} + (1 - S_{lr} - S_{gr})S_e, \tag{1.3}$$

and the corresponding Mualem–van Genuchten relative-permeability law uses the lower cutoff $k_{\min}$:

$$k_{\mathrm{rel}}(S_l) = \max\left(k_{\min}, S_e^{1/2}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2\right). \tag{1.4}$$

This closure follows the standard van Genuchten and Mualem forms [23, p. 892, Eqs. 2–3 and p. 893, Eq. 8] [16, pp. 513–515, Eqs. 1–2, 14, and 21]. In the present workflow these retention and relative-permeability parameters are fixed model structure, not released calibration fields; their values are declared in `04_2_media_twophase.xml`.

Table 1.4: Parameters introduced by the Darcy, retention, and relative-permeability equations.

XML name or law	Symbol	Active value	Role in Equations (1.1)–(1.4)
`k_i/k_i_rd`	$\boldsymbol{K}$	Baseline tensor $\begin{bmatrix}1.0 & 0\\ 0 & 0.4\end{bmatrix} 10^{-19}\,\mathrm{m}^2$; current OGS mesh field varies magnitude	Intrinsic permeability tensor in Darcy velocity.
`eta`	μ_l	$1.0 \times 10^{-3}\,\mathrm{Pa\,s}$	Liquid viscosity in Darcy mobility.
VG saturation law	S_{lr}, S_{gr}, m, p_b	$0.1, 0, 0.45, 10\,\mathrm{MPa}$	Retention curve $S_l(p_c)$ with $p_c = -p_l$.
VG relative permeability	k_{rel}, $k_{\min}$	same residual saturations and $m = 0.45$, with $k_{\min} = 10^{-25}$	Unsaturated mobility reduction multiplying $\boldsymbol{K}/\mu_l$.

1.3.3 Liquid Mass Balance

The no-vapor liquid mass equation can be written as follows. Overdots denote time derivatives. The coefficient β_l is the pressure-derivative slot associated with the liquid-density law, α_B is the Biot coefficient, α_T^s is the solid thermal-expansivity parameter, and Q_H is the hydraulic source slot:

$$\left(S_l\beta_l - n\frac{\partial S_l}{\partial p_c}\right)\rho_l\,\dot{p}_l - S_l\left(\frac{\partial \rho_l}{\partial T} + \rho_l(\alpha_B - S_l)\alpha_T^s\right)\dot{T} + \nabla\cdot(\rho_l\boldsymbol{v}_l) + S_l\alpha_B\rho_l\,\nabla\cdot\dot{\boldsymbol{u}} = Q_H. \tag{1.5}$$

The equation shows why permeability, retention, porosity, density, temperature, and mechanical volumetric strain cannot be interpreted independently. In particular, changing the boundary pressure curve or the retention law can mimic a change in permeability through the desaturation timing. The active XML does not add a separate hydraulic source term, so Q_H is retained only as the standard equation slot [19, "Governing equations without vapor diffusion"].

The liquid density is pressure- and temperature-dependent:

$$\rho_l(T, p_l) = 1095\left[1 - 5\times 10^{-4}(T - 298.15) + 3.2\times 10^{-10}(p_l - 7.5\times 10^6)\right]. \tag{1.6}$$

Thus pressure storage, saturation storage, thermal density coupling, and mechanical volumetric coupling belong to the full TRM statement. Under the current active temperature constraints, the temperature-dependent parts reduce as stated in Section 1.3.6.

Table 1.5: Parameters introduced by the liquid mass balance and density law.

XML name or law	Symbol	Active value	Role in Equations (1.5)–(1.6)
`phi/n_rd`	n	0.105 in the current field package	Porosity in liquid storage, retention derivative, thermal mixing, and $\theta = nS_l$.
liquid density law	$\rho_l(T, p_l)$	$1095[1-5\times10^{-4}(T-298.15)+3.2\times10^{-10}(p_l - 7.5\times10^6)]$	Full TRM density law. Under the active isothermal constraint it reduces to $\rho_l^0(p_l)$ in Equation (1.13).
pressure slope in liquid density law	β_l	$3.2\times10^{-10}\,\mathrm{Pa}^{-1}$ relative pressure slope in the linear density law	Used in the pressure-storage slot of Equation (1.5); not a separately released calibration parameter.
`biot`	α_B	1	Couples volumetric strain and pore pressure to hydraulic storage and mechanics.
solid thermal expansivity	α_T^s	`CTE`=1254.74 in XML	Appears in the thermal-coupling term of Equation (1.5); value and unit remain the CTE provenance caveat recorded in Table 1.7.

XML name or law	Symbol	Active value	Role
hydraulic source	Q_H	No separate active source in the process file	Retained as the standard equation slot, but not used as an additional forcing term.

1.3.4 Quasi-static Mechanics

The mechanical balance is quasi-static. The vector $\boldsymbol{f}$ is the body-force term, $b(S_l)$ is the Bishop saturation factor, and α_B is the Biot coefficient:

$$\nabla\cdot(\boldsymbol{\sigma} - b(S_l)\alpha_B p_l \boldsymbol{I}) + \boldsymbol{f} = \boldsymbol{0}. \tag{1.7}$$

The Bishop stress factor $b(S_l)$ represents the saturation-dependent weighting of the pore-pressure term in unsaturated effective stress [4, bibliographic record, no DOI found]. For the reference setup, the OGS Bishop law has exponent one and the Biot coefficient is one, so $b(S_l) = S_l$ and $\alpha_B = 1$ after substitution [19, TRM momentum equation and media-property table]; the same values are declared by `BishopsPowerLaw` and `biot` entries in `04_2_media_twophase.xml` and `03_parameters_TRM.xml`. We keep the symbolic form in (1.7) because it makes the coupling assumptions explicit.

The effective stress follows orthotropic linear elasticity with thermal and swelling strain contributions:

$$\boldsymbol{\sigma} = \mathbb{C} : \left(\boldsymbol{\varepsilon}(\boldsymbol{u}) - \boldsymbol{\varepsilon}^{\mathrm{th}}(T) - \boldsymbol{\varepsilon}^{\mathrm{sw}}(S_l)\right), \qquad \boldsymbol{\varepsilon}(\boldsymbol{u}) = \frac{1}{2}\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\top}\right). \tag{1.8}$$

This is not a staged excavation model. The current mechanics model starts from the already open two-dimensional geometry and has no active initial stress tensor or active top load. Any future mechanical residual therefore requires a separate decision about prestress, excavation unloading, and the reference displacement state. This follows directly from `01_processes_TRM.xml`, where `initial_stress` is commented out, and from `02_process_variables_TRM.xml`, where the top-load Neumann block is commented out.

Table 1.6: Parameters introduced by the quasi-static mechanics equations.

XML name or law	Symbol	Active value	Role in Equations (1.7)–(1.8)
`specific_body_force`	$\boldsymbol{f}$	$(0, 0)$	Body force is disabled in the momentum equation.
`biot`	α_B	1	Pore-pressure coupling in the effective stress term.
`BishopsPowerLaw`	$b(S_l)$	S_l	Effective-stress scaling because the configured exponent is one.
`E`	E_i	$(13.8, 6.0, 13.8)$ GPa	Orthotropic Young moduli in $\mathbb{C}$.
`G`	G_i	$(4.79, 2.46, 4.79)$ GPa	Orthotropic shear moduli in $\mathbb{C}$.
`nu`	ν_i	$(0.44, 0.22, 0.44)$	Orthotropic Poisson ratios in $\mathbb{C}$.
swelling stress rate	p_{sw}	1.0 MPa in each direction	Saturation-dependent swelling law with exponents one and limits $0.1 \le S_l \le 1$.
`ic_sigma0`, `load_top`	$\boldsymbol{\sigma}_0$, top load	Defined but inactive	Prestress and top traction are provenance entries, not active mechanical inputs.

1.3.5 Heat Balance

Let $(\rho c_p)^{\mathrm{eff}}$ be effective heat storage and $\boldsymbol{k}_T^{\mathrm{eff}}$ the effective thermal-conductivity tensor. With liquid heat capacity c_l and thermal source slot Q_T, the TRM heat balance is

$$(\rho c_p)^{\mathrm{eff}}\dot{T} - \nabla\cdot\left(\boldsymbol{k}_T^{\mathrm{eff}}\nabla T\right) + \rho_l c_l\,\nabla T \cdot \boldsymbol{v}_l = Q_T. \tag{1.9}$$

The effective heat storage and thermal conductivity use porosity and saturation mixing:

$$(\rho c_p)^{\mathrm{eff}} = (1-n)\rho_s c_s + n S_l \rho_l c_l, \qquad \boldsymbol{k}_T^{\mathrm{eff}} = (1-n)k_s\boldsymbol{I} + n S_l k_l \boldsymbol{I}. \tag{1.10}$$

The active temperature initial and boundary values are both 298.15 K. In the checked XML, the temperature Dirichlet condition is attached to `bulk_all`; `cd_a_open_niche_quad.prj` lists `bulk_all.vtu` as a project mesh. This boundary support is therefore read as the domain-wide temperature constraint used by the active run. Thermal parameters are therefore fixed coupling context rather than released thermal calibration fields. The `CTE`=1254.74 XML value remains a provenance caveat because it is numerically identical to `c_p_s` and carries a heat-capacity-like XML comment in `03_parameters_TRM.xml`.

Table 1.7: Parameters introduced by the heat balance and thermal mixing equations.

XML name or law	Symbol	Active value	Role in Equations (1.9)–(1.10)
`rho_s`	ρ_s	$2.52 \times 10^3\,\mathrm{kg\,m^{-3}}$	Solid density in effective heat storage.
`c_p_s`	c_s	$1254.74\,\mathrm{J\,kg^{-1}K^{-1}}$	Solid heat capacity in effective heat storage.
`kappa_th_s`	k_s	$2.156\,\mathrm{W\,m^{-1}K^{-1}}$	Solid thermal conductivity in effective mixing.
`c_p_l`	c_l	$4160\,\mathrm{J\,kg^{-1}K^{-1}}$	Liquid heat capacity in storage and advective heat transport.
`kappa_l`	k_l	$0.6\,\mathrm{W\,m^{-1}K^{-1}}$	Liquid thermal conductivity in effective mixing.
`CTE`	α_T^s	1254.74 in XML	Thermal-expansivity parameter used by OGS; value and unit remain a provenance caveat.
thermal source	Q_T	No separate active source in the process file	Retained as the standard equation slot, but not used as an additional forcing term.

1.3.6 Isothermal Reduced Model for the Active Parametrization

The active OGS process is still TRM, and T is still a primary variable in the project file. However, the active initial and boundary data prescribe

$$T(\boldsymbol{x}, t) = T_0 := 298.15\,\mathrm{K} \quad \text{for all } \boldsymbol{x} \in \Omega,\ t \in I. \tag{1.11}$$

Therefore

$$\dot{T} = 0, \qquad \nabla T = \mathbf{0}, \qquad \boldsymbol{\varepsilon}^{\mathrm{th}}(T) - \boldsymbol{\varepsilon}^{\mathrm{th}}(T_0) = \mathbf{0}. \tag{1.12}$$

Flow and displacement changes do not generate temperature changes in this active model, because the temperature degree of freedom is constrained. The heat equation is solved by the constant field and does not add an independent evolution equation for the present parametrization [19, TRM heat equation and process-variable list].

For interpretation and parameter fitting, the active model can therefore be read as an isothermal Richards–mechanics model with primary unknowns $(\boldsymbol{u}, p_l)$. The liquid density reduces from (1.6) to

$$\rho_l^0(p_l) = 1095 \left[1 + 3.2 \times 10^{-10}(p_l - 7.5 \times 10^6)\right], \tag{1.13}$$

so density can still vary with pressure, but not with temperature. The liquid mass balance becomes

$$\left(S_l \beta_l - n \frac{\partial S_l}{\partial p_c}\right) \rho_l^0\, \dot{p}_l + \nabla \cdot (\rho_l^0 \boldsymbol{v}_l) + S_l \alpha_B \rho_l^0\, \nabla \cdot \dot{\boldsymbol{u}} = Q_H, \tag{1.14}$$

with the same Darcy velocity (1.1), retention law (1.3), and relative-permeability law (1.4). The mechanical balance keeps the same pore-pressure coupling, while the thermal-strain increment drops out:

$$\nabla \cdot (\boldsymbol{\sigma} - b(S_l)\alpha_B p_l \boldsymbol{I}) + \boldsymbol{f} = \mathbf{0}, \qquad \boldsymbol{\sigma} = \mathbb{C} : (\boldsymbol{\varepsilon}(\boldsymbol{u}) - \boldsymbol{\varepsilon}^{\mathrm{sw}}(S_l)). \tag{1.15}$$

This reduced form is not a different OGS project. It is the mathematical model that the current TRM project realizes once the homogeneous constant temperature constraint is applied.

1.3.7 Mathematical Boundary and Initial Data

Let the outer square boundary be

$$\Gamma_\square := \Gamma_A \cup \Gamma_B \cup \Gamma_C \cup \Gamma_D,$$

and let Γ_E denote the open-niche boundary. The reference mathematical hydraulic boundary inventory is

$$p_l = p_\square = 1.5 \times 10^6\,\mathrm{Pa} \quad \text{on } \Gamma_\square, \qquad p_l = p_E(t) \quad \text{on } \Gamma_E. \tag{1.16}$$

Thus the reference mathematical boundary inventory uses a Dirichlet hydraulic condition on all four sides of the square. The niche boundary is the only time-dependent hydraulic Dirichlet boundary. Figure 1.1 plots the tabulated $p_E(t)$ data read from `08_08_open_niche_seasonal.xml`. This is the model-time boundary-data vector used by the current project, not the commented shifted calendar-time variant. The negative values of $p_E(t)$ are interpreted through (1.2): they are liquid-pressure values corresponding to positive suction [30, Sec. 3, Eqs. 1–2].

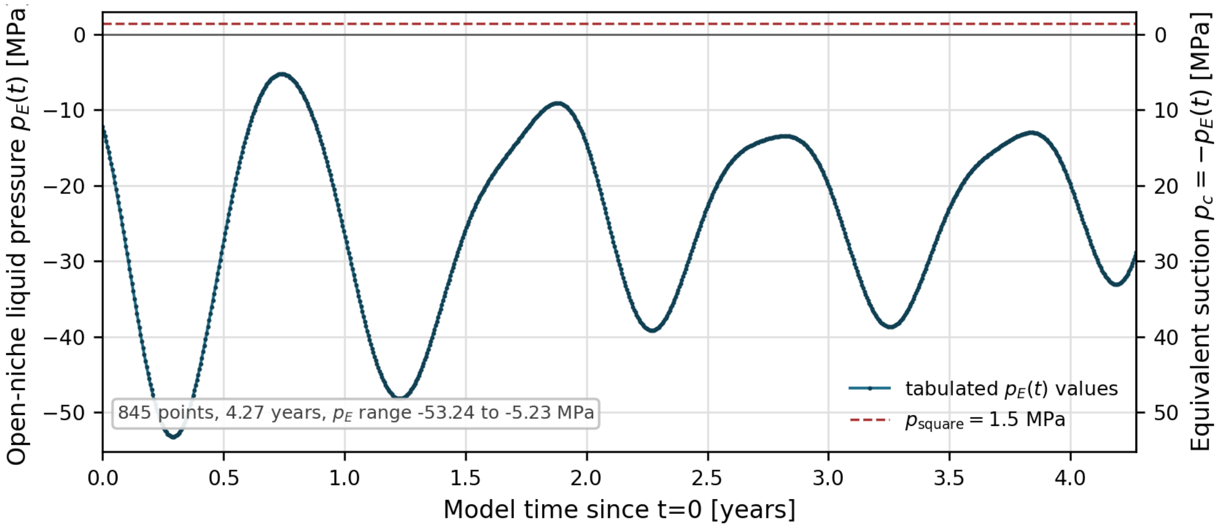

Figure 1.1: Tabulated open-niche pressure boundary data $p_E(t)$ prescribed on Γ_E. The markers are the 845 XML values used by the active OGS setup. The left axis shows liquid pressure; the right axis shows the equivalent suction under $p_c = -p_E(t)$. The dashed line marks the constant square-boundary pressure $p_\square = 1.5\,\mathrm{MPa}$ for scale.

The figure is an input diagnostic, not a measurement-provenance closure. It shows the numerical forcing passed to OGS. The source sensors, filtering, time-origin convention, and uncertainty of this boundary-data vector remain the boundary-input gate recorded in Section 1.5.

The mechanical boundary inventory is component-wise kinematic support on the outer square,

$$u_x = 0 \quad \text{on } \Gamma_A \cup \Gamma_B, \qquad u_y = 0 \quad \text{on } \Gamma_C \cup \Gamma_D, \tag{1.17}$$

with natural tangential components and natural niche traction in the current model. The thermal boundary is $T = 298.15\,\mathrm{K}$ on the thermal support used by the OGS setup. The initial conditions are

$$\boldsymbol{u}(\boldsymbol{x}, 0) = \boldsymbol{0}, \qquad p_l(\boldsymbol{x}, 0) = 1.5 \times 10^6\,\mathrm{Pa}, \qquad T(\boldsymbol{x}, 0) = 298.15\,\mathrm{K}. \tag{1.18}$$

The pressure expression in the XML contains a vertical term multiplied by zero. Consequently, the active initial pressure field is spatially uniform and does not encode a drained near-niche post-excavation pressure field.

1.4 OGS System Realization

1.4.1 Project Blocks and Process Variables

The OGS realization is the `THERMO_RICHARDS_MECHANICS` process in `01_processes_TRM.xml`. The project assigns polynomial order 2 to displacement and order 1 to pressure and temperature. The same process block exposes `velocity`, `saturation`, `sigma`, `epsilon`, and `swelling_stress` as secondary variables. Gravity is off because `specific_body_force` is `0 0` [19, "Process variables"] [18, initial and boundary-condition structure].

The time-loop files bind this process to Backward Euler time discretization, fixed time stepping, and the nonlinear solver `basic_newton`. The nonlinear solver is a Newton solver with at most 150 iterations and delegates linear systems to `general_linear_solver`, which uses `PardisoLU`. The convergence criterion is component-wise Δx in the L^2 norm, with absolute tolerance 10^{-10} and relative tolerance 10^{-8} for each primary-variable component. These settings define the numerical realization of the frozen TRM equations; they do not change the governing equations or the released material-field policy.

Table 1.8 maps the mathematical quantities to the OGS blocks, parameters, mesh fields, and output arrays. This table is the bridge between the mathematical model and the file-level system. It also prevents us from treating an observation operator, such as NMR water content or ERT resistivity, as if it were an OGS primary variable.

Table 1.8: Mapping between mathematical quantities and the OGS realization.

Mathematical quantity	OGS block, tag, or field	Input realization	Output or downstream use
$\boldsymbol{u}$	Process variable `displacement` in `02_process_variables_TRM.xml`.	Two components, polynomial order 2, initial parameter `ic_displacement`; component-wise Dirichlet supports on the outer square.	Current OGS VTU array `displacement`. Mechanical observations need a reference-zero and support operator.
p_l	Process variable `pressure`.	One component, polynomial order 1, initial parameter `ic_pressure`; Dirichlet pressure data on the reference hydraulic supports.	Current OGS VTU array `pressure`. Pressure measurements and RH-derived suction require sign and reference-pressure conventions.
T	Process variable `temperature`.	One component, polynomial order 1, initial parameter `ic_temperature`; Dirichlet temperature parameter `bc_temperature_outside`.	Current OGS VTU array `temperature`. In the active parametrization it is constrained to $T_0 = 298.15\,\mathrm{K}$, so thermal gradients and temperature-change terms vanish.
$\boldsymbol{K}$	Medium property `permeability`, parameter `k_i`.	Current OGS `MeshElement` field `k_i_rd` on the projected bulk mesh. The source package also contains a constant tensor fallback.	Not a dynamic state variable. Observation operators sample or average the mesh field for pulse-test permeability comparisons.
n	Medium property `porosity`, parameter `phi`.	Current OGS `MeshElement` field `n_rd`; current field package is spatially constant at 0.105.	Current OGS VTU array `porosity`. Used with saturation to form $\theta = nS_l$.
S_l	Medium property `saturation` and process secondary variable.	Computed by `SaturationVanGenuchten` from $p_c = -p_l$; not prescribed as a primary variable.	Current OGS VTU array `saturation`. Main state field for water-content observation operators.
k_{rel}	Medium property `relative_permeability`.	Computed by the OGS van Genuchten relative-permeability law; fixed parameters in the reference setup.	Usually not exported as a required VTU field. It affects pressure and saturation through Darcy mobility.

Mathematical quantity	OGS block, tag, or field	Input realization	Output or downstream use
$\boldsymbol{v}_l$ and $\boldsymbol{q}_l$	Secondary variable `velocity`.	Computed from $\boldsymbol{K}$, k_{rel}, μ_l, and ∇p_l.	Available from the process if requested. Flux residuals require an explicit Darcy-velocity versus pore-velocity convention.
$\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$, swelling stress	Secondary variables `sigma`, `epsilon`, and `swelling_stress`.	Computed from the orthotropic elasticity law, Bishop/Biot coupling, thermal strain, and saturation-dependent swelling.	Available from the process if requested. Current HM measurements remain gated by prestress, excavation, and reference-zero decisions.
$p_E(t)$	`open_niche_seasonal` and `open_niche_seasonal_curve`.	`CurveScaled` parameter with unit scaling factor 1.0.	Active boundary forcing in the reference setup. Its measurement provenance, filtering, and uncertainty remain open.

1.4.2 Mesh and Boundary Supports

The transferred project uses boundary submeshes `cd-a_left`, `cd-a_right`, `cd-a_top`, `cd-a_bottom`, and `cd-a_niche4`. Figure 1.2 gives the schematic notation for these supports. The figure is only a notation aid; the boundary-condition table below gives the model-facing mapping to XML tags and prescribed data.

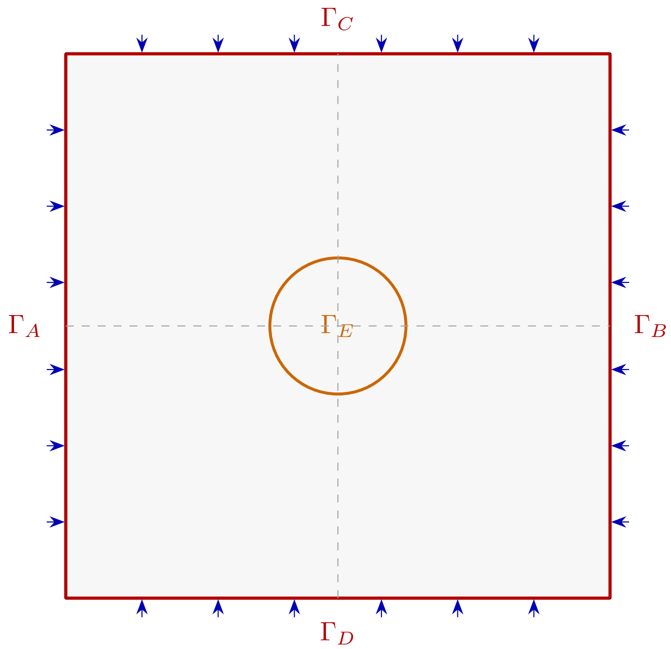

Figure 1.2: Schematic boundary-condition map for the reference two-dimensional model. The outer square boundary is denoted by $\Gamma_A \cup \Gamma_B \cup \Gamma_C \cup \Gamma_D$, and the niche boundary is denoted by Γ_E.

Table 1.9: Boundary-set notation, XML tags, and prescribed mechanical/hydraulic data.

Boundary set	XML tag name and parameter	Mechanical BC	Hydraulic BC
Γ_A	`cd-a_left`; `bc_displacement`; reference square-pressure parameter `bc_pressure_outside`.	Dirichlet $u_x = 0$; tangential component natural.	Dirichlet $p_l = p_\square$, $p_\square = 1.5 \times 10^6$ Pa.
Γ_B	`cd-a_right`; `bc_displacement`; reference square-pressure parameter `bc_pressure_outside`.	Dirichlet $u_x = 0$; tangential component natural.	Dirichlet $p_l = p_\square$, $p_\square = 1.5 \times 10^6$ Pa.
Γ_C	`cd-a_top`; `bc_displacement`; `bc_top_pressure`.	Dirichlet $u_y = 0$; tangential component natural; `load_top` inactive.	Dirichlet $p_l = p_\square$, $p_\square = 1.5 \times 10^6$ Pa.
Γ_D	`cd-a_bottom`; `bc_displacement`; reference square-pressure parameter `bc_pressure_outside`.	Dirichlet $u_y = 0$; tangential component natural.	Dirichlet $p_l = p_\square$, $p_\square = 1.5 \times 10^6$ Pa.
$\Gamma_A \cup \Gamma_B$	`cd-a_left`, `cd-a_right`.	Compact mechanics notation: $u_x = 0$.	Part of the square hydraulic Dirichlet boundary $p_l = p_\square$.
$\Gamma_C \cup \Gamma_D$	`cd-a_top`, `cd-a_bottom`.	Compact mechanics notation: $u_y = 0$.	Part of the square hydraulic Dirichlet boundary $p_l = p_\square$.
$\Gamma_\square = \Gamma_A \cup \Gamma_B \cup \Gamma_C \cup \Gamma_D$	`cd-a_left`, `cd-a_right`, `cd-a_top`, `cd-a_bottom`.	Outer-box kinematic support, component-wise.	Dirichlet $p_l = p_\square = 1.5 \times 10^6$ Pa on all four sides of the outer square in the reference boundary inventory. Checked OGS XML attachment for left/right/bottom remains a consistency gate.
Γ_E	`cd-a_niche4`; `open_niche_seasonal`; `open_niche_seasonal_curve`; `pressure_scaling_factor`.	No prescribed displacement or traction in the active boundary-condition block.	Dirichlet $p_l = p_E(t)$. The curve has 845 tabulated values, spans 0 to 1.3482×10^8 s, and ranges from about -5.23 to -53.24 MPa.

The table states the required reference boundary inventory: constant hydraulic Dirichlet data on all four sides of the square and a time-dependent niche pressure curve. The implementation gate is file-level: in the checked `02_process_variables_TRM.xml`, explicit pressure Dirichlet conditions are attached to `cd-a_top` and `cd-a_niche4`. Therefore the left, right, and bottom square-pressure conditions must either be attached in the reference OGS setup or documented as a preprocessing/generated-project step. This gate does not change the mathematical boundary inventory in (1.16); it states what the implementation must prove.

1.4.3 Material Fields and OGS Inputs

The OGS parameter block can define constants, functions, curve-scaled data, and mesh element or mesh node fields [17, sections "CurveScaled", "Function", and "MeshElement and MeshNode"]. The current OGS workflow uses `MeshElement` parameters for intrinsic permeability and porosity. The intrinsic permeability tensor is supplied through `k_i_rd`; this is the active material field whose magnitude is varied in the present inversion workflow. Porosity is supplied through `n_rd`, but in the current field package all cells have value 0.105. This is a mesh-field representation of fixed support, not a released porosity inversion.

Table 1.10: Material and constitutive fields in the active model or tied to OGS fields.

Field	Material meaning	Relation to other fields	OGS role in the current workflow
Intrinsic permeability tensor $\boldsymbol{K}(\boldsymbol{x})$	Clay/EDZ effective Darcy mobility before relative permeability and viscosity are applied. It is the main hydraulic material field for the present inversion.	Hydraulic conductivity is derived from $\boldsymbol{K}$, liquid density, viscosity, relative permeability, and saturation state; it is not an independent base field here.	Current OGS `MeshElement` field `k_i_rd`. Current release gates allow permeability magnitude changes while tensor shape remains staged.
Porosity $n(\boldsymbol{x})$	Effective pore-volume fraction used by the OGS continuum model.	Water content is $\theta = nS_l$ only for the mobile/OGS liquid storage; NMR, ERT, and Taupe/TDR may also see bound/interlayer water or calibration offsets.	Current OGS `MeshElement` field `n_rd`, fixed at 0.105 in the current field package. Not released as an unknown yet.

Field	Material meaning	Relation to other fields	OGS role in the current workflow
Saturation S_l and retention $S_l(p_c)$	Liquid occupancy of the OGS pore space, governed by a retention law.	$p_c = -p_l$ under the active gas/reference convention; the same law controls storage derivative and water-content output.	Fixed OGS van Genuchten saturation law with $S_{lr} = 0.1$, $S_{gr} = 0$, exponent $m = 0.45$, and $p_b = 10\,\mathrm{MPa}$.
Relative permeability $k_{\mathrm{rel}}(S_l)$	Reduction of liquid mobility in unsaturated conditions.	Multiplies $\boldsymbol{K}/\mu_l$ in Darcy velocity; should not be fitted independently of retention without a release decision.	Fixed OGS van Genuchten relative-permeability law with the same residual saturations, exponent 0.45, and minimum 10^{-25}.
Biot and Bishop coupling	Poromechanical coupling between pore pressure, saturation, and stress.	$\alpha_B = 1$ and $b(S_l) = S_l$ reduce the pore-stress scaling to $S_l p_l \boldsymbol{I}$.	Fixed XML parameter `biot`=1 and `BishopsPowerLaw` exponent 1.
Orthotropic elasticity $\mathbb{C}$	Directional stiffness of Opalinus Clay in the 2D mechanical approximation.	Controls stress/displacement response together with pore pressure, saturation, swelling, and any prestress assumption.	Fixed orthotropic linear elasticity: $E = (13.8, 6.0, 13.8)$ GPa, $G = (4.79, 2.46, 4.79)$ GPa, $\nu = (0.44, 0.22, 0.44)$. Mechanical release is gated.
Swelling stress/strain	Saturation-driven mechanical source representing swelling response.	Coupled to saturation and constrained by mechanical boundary assumptions; can mimic deformation from pressure changes if released prematurely.	Active fixed saturation-dependent swelling law with 1.0 MPa swelling pressures in all directions and limits $S_l \in [0.1, 1]$.
Thermal and liquid properties	Solid/liquid storage, conductivity, density, viscosity, and isothermal coupling context.	In the full XML, liquid density depends linearly on T and p_l, and thermal expansion can enter THM coupling. In the active reduction, $T = T_0$, so only the pressure part of density varies.	Fixed XML constants/laws. The `CTE`=1254.74 value is a provenance caveat and should not be interpreted as a calibrated thermal-expansion coefficient.
Fault, fracture, crack, and EDZ fields	Structural features that may localize permeability, porosity, stiffness, or aperture.	Measured aperture could imply hydraulic aperture or localized permeability only after scale, clay-contact, and 3D-to-2D projection assumptions are accepted.	Not an active OGS geometry/material class in this chapter. They remain structural priors, masks, scenarios, or qualitative diagnostics until source data are resolved.
Time-dependent material fields	Possible healing, sealing, fracture closure, or EDZ evolution over the multi-year experiment.	Could appear as $K(\boldsymbol{x}, t)$, $n(\boldsymbol{x}, t)$, retention changes, or boundary/input changes, but may be confounded with forcing and saturation state.	Excluded from the first static permeability-field workflow. Release requires same-support repeated evidence and state-output diagnostics.

The material-field question is therefore not only which constants appear in XML. For each field we need to know its material meaning, whether it is a base field or a derived field, whether it is independent enough to release, and how OGS consumes it. In the first workflow, $\boldsymbol{K}(\boldsymbol{x})$ is the only released base material field. Hydraulic conductivity, water content, storativity, resistivity, and flux are derived from $\boldsymbol{K}$, n, S_l, fluid properties, and observation operators [19, Darcy law and effective-storage equation] [30, Sec. 3, Eq. 3].

1.4.4 Initial Conditions, Time Grid, and Outputs

The initial-condition XML entries realize (1.18). The time-step file comments identify $t = 0$ with 18 September 2019. The active grid starts at zero seconds, uses fixed Backward Euler stepping, and ends at 1.4×10^8 s. The first block uses 365 steps of 864,000 s, and the second block uses 365 steps of 1,314,900 s. The output schedule is intended to match the monthly ERT cadence after the first output, as stated in comments in `05_time_loop_TRM.xml` and `05_1_fixed_timestepping.xml`.

Table 1.11: Initial conditions used by the active TRM process.

Variable	XML parameter	Active value	Interpretation and modelling gate
$\boldsymbol{u}(\boldsymbol{x}, 0)$	`ic_displacement`	$(0, 0)$ m	The model starts from zero displacement on the already open two-dimensional geometry. No excavation/driving phase is represented explicitly.
$p_l(\boldsymbol{x}, 0)$	`ic_pressure`	1.5×10^6 Pa everywhere	The XML function contains a vertical term multiplied by zero, therefore the active initial liquid pressure is spatially uniform. This remains a gate for pressure history: no post-excavation drained niche field is encoded.
$T(\boldsymbol{x}, 0)$	`ic_temperature`	298.15 K	The model is initialized at T_0. Together with the `bulk_all` temperature Dirichlet condition, this enforces homogeneous constant temperature in the active parametrization.
$\boldsymbol{\sigma}_0$	`ic_sigma0`	Defined but inactive	The file defines in-situ stress-like expressions, but the process XML comments out the initial-stress reference. Prestress is therefore a provenance question, not an active initial condition in the present run.

This initial state is a numerical post-excavation state. It is not a simulation of the initial excavation or gradual niche unloading. In mechanics, `ic_sigma0` is defined but not connected as active initial stress, and `load_top` remains inside a commented top-load block. In hydraulics, the initial pressure is uniform and does not reconstruct near-niche drainage from nearby holes, RH/suction, or direct pressure measurements. Early transients may therefore mix physical drainage with adjustment from the prescribed initial state.

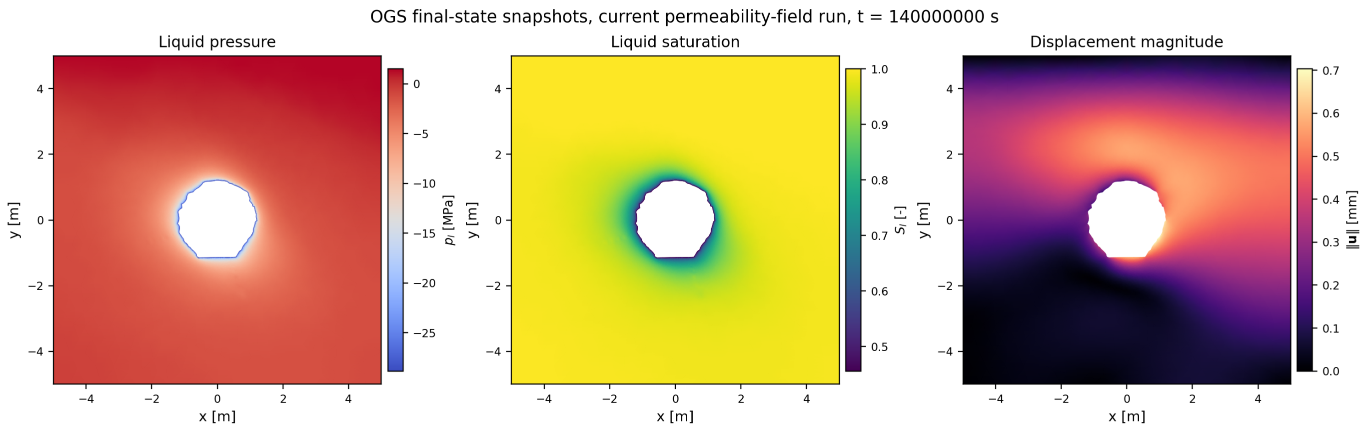

Figure 1.3: Final-state snapshots from the current permeability-field OGS run at $t = 1.4 \times 10^8$ s. The panels show liquid pressure, liquid saturation, and displacement magnitude from the current OGS VTU output. The figure is a diagnostic visualization of the active solver state; it is not itself a measurement residual.

1.4.5 Model Outputs and Observation Operators

The OGS output layer must stay separate from measurement-specific observation operators. The current OGS workflow requests pressure, saturation, temperature, displacement, and porosity. These arrays are model states or fixed support fields, not residuals by themselves. A residual exists only after an observation operator defines the measured quantity, support, time mapping, unit convention, reference state, and uncertainty. Permeability comparisons are a separate case: they sample or average the input tensor field, because permeability is not a dynamic state output.

Table 1.12: Model outputs and observation-facing quantities.

Quantity	OGS status	Measurement tie	Required operator or gate
p_l	Primary variable and current OGS VTU output.	Direct pressure or mini-piezometer data, pressure-boundary checks, and pressure diagnostics.	Reference convention, elevation/head conversion, gauge/absolute convention, and 2D projection error must be explicit.
$\boldsymbol{u}$	Primary variable and current OGS VTU output.	Geoscope, crackometer, levelling, extensometer, laser-scan, and deformation screens.	Reference-zero, sign convention, support geometry, and whether displacement can be used as a hard residual are unresolved.
T	Primary variable and current OGS VTU output.	Constrained field for the present isothermal workflow.	Thermal residuals are not active; $T = T_0$ fixes thermal gradients and temperature-change terms.
S_l	Secondary variable and current OGS VTU output.	Water-content streams after transformation: NMR, Taupe/TDR, and ERT-derived moisture.	Must be converted to $\theta = nS_l$ or another accepted observable; bound-water and calibration policies remain gates.
n	Material field and current OGS VTU output.	Support for $\theta = nS_l$, not an observed active unknown yet.	Current $n = 0.105$ support must be retained unless NMR/ERT/Taupe policies justify a porosity release.
$\boldsymbol{K}$ or $e^\top \boldsymbol{K} e$	Input field, not a state output unless sampled from mesh data.	Pulse-test permeability/transmissibility interpretations.	Interval orientation, support averaging, gas/slip correction, and duplicate-support likelihood policy are required.
$\boldsymbol{v}_l$, $\boldsymbol{q}_l$	Secondary velocity available from OGS; flux can be derived.	Hydraulic diagnostics and possible flow-rate checks if such data are supplied.	Use Darcy velocity, not pore velocity, unless an observation operator explicitly requires division by nS_l.
$\boldsymbol{\sigma}, \boldsymbol{\varepsilon}$, swelling stress	Secondary variables available from the process.	Mechanical plausibility, crack/deformation interpretation, and the prestress gate.	Initial stress and excavation unloading are not active; mechanical conclusions need separate staged-model or post-excavation assumptions.
$\theta = nS_l$	Post-processing product.	NMR and Taupe/TDR water-content comparison; ERT if transformed through a water-content/resistivity relation.	Decide absolute versus bias-corrected versus trend/anomaly comparison and handle bound/interlayer water.
Resistivity or log-resistivity proxy	Observation-model product, not an OGS state variable.	ERT inversion fields.	Requires Archie/clay-specific transform, pore-water conductivity/salinity treatment, surface conduction, anisotropy, support aggregation, and covariance policy.

1.5 Model-facing Question Inventory and Proposed Answers

This section keeps only the questions that define the model itself: boundary conditions, initial conditions, OGS inputs, OGS outputs, and material or constitutive field descriptions. Table 1.13 keeps those questions granular. The inventory is split into model-facing subtables; each item is written as a shaded question row followed by an unshaded answer row. The status colour in the answer row tells how the answer may be used. Established, checked, and fixed rows can be used as the current model description. Gate and caveat rows identify the exact question whose answer is not strong enough for hard calibration, validation, or model release.

Table 1.13: Granular model-facing questions and proposed answers from the checked OGS project files and cited sources. Questions are shaded light grey; answers are unshaded and carry a coloured status label.

Answer-status legend.

Established	Directly supported by the checked OGS project files, equations, published papers, or official OGS documentation.
Checked input	Numerically or file-level checked, but not necessarily supported as a full measurement provenance statement.
Fixed context	Active or inactive context is known and should be treated as fixed in the current baseline.
Release policy	Current modelling or inversion-policy decision, not a material fact by itself.
Gate	Partly answered, but external evidence, source recovery, or collaborator decision is required before hard use.
Caveat	Known limitation, inactive entry, or provenance problem that must not be promoted to calibrated physics.

Hydraulic Boundary Conditions

Question. Which mathematical pressure boundary inventory should the model state?	
Established	Use $p_l = p_\square = 1.5\,\mathrm{MPa}$ on $\Gamma_\square = \Gamma_A \cup \Gamma_B \cup \Gamma_C \cup \Gamma_D$, and $p_l = p_E(t)$ on the niche boundary Γ_E.
Question. Which pressure Dirichlet conditions are attached in the checked OGS XML?	
Gate	The checked `02_process_variables_TRM.xml` attaches pressure Dirichlet data to `cd-a_top` through `bc_top_pressure` and to `cd-a_niche4` through `open_niche_seasonal`. It does not explicitly attach left/right/bottom pressure boundaries in that file, so the implementation remains a consistency gate.
Question. Are the outer-square pressure boundaries time-dependent?	
Established	No. The square-boundary pressure is the constant value $p_\square = 1.5\,\mathrm{MPa}$. The only time-dependent hydraulic boundary in the current pressure inventory is the open-niche curve $p_E(t)$.
Question. What does the negative niche-pressure curve mean?	
Established	The values are OGS liquid-pressure values in the Richards sign convention. Because $p_c = -p_l$, negative p_l represents positive suction/capillary pressure, not a separate pressure variable.
Question. What is known about the active niche curve $p_E(t)$?	
Checked input	`open_niche_seasonal` is a `CurveScaled` parameter with unit scaling factor. The curve has 845 tabulated values, spans 0 to 1.3482×10^8 s, and ranges from about -5.23 to $-53.24\,\mathrm{MPa}$.
Question. What remains unknown about the active niche curve $p_E(t)$?	
Gate	The checked OGS project gives the active curve values, but the cited source boundary does not close the source sensors, source columns, filtering, smoothing, interpolation, time origin, and uncertainty for the active curve.

Mechanical Boundary Conditions

Question. Which displacement constraints are active on the square boundary?	
Established	The model fixes $u_x = 0$ on $\Gamma_A \cup \Gamma_B$ through `cd-a_left`/`cd-a_right`, and fixes $u_y = 0$ on $\Gamma_C \cup \Gamma_D$ through `cd-a_top`/`cd-a_bottom`.
Question. What mechanical condition is used on the niche boundary?	
Established	No displacement or traction condition is prescribed on Γ_E in the active boundary-condition block. The niche is therefore natural mechanically in the current formulation.
Question. Is the top mechanical load active?	
Caveat	No. The `load_top` parameter is defined in the parameter file, but the Neumann top-load boundary block is commented out in the process-variable XML.

Thermal Boundary And Reduced Temperature Model

Question. Which temperature boundary and initial value are active?	
Fixed context	$T = 298.15\,\mathrm{K}$ is prescribed on `bulk_all`, and $T(\boldsymbol{x}, 0) = 298.15\,\mathrm{K}$. This constrains temperature to be homogeneous and constant in the active model, although the TRM heat equation remains present in the project.

Initial State And Excavation Interpretation

Question. What is the active initial pressure field?	
Established	`ic_pressure` is a function, but its vertical term is multiplied by zero. The active initial liquid pressure is spatially uniform 1.5 MPa.
Question. Does the initial pressure encode post-excavation drainage near the niche?	
Gate	No. The current input has no heterogeneous initial drainage field from RH/suction, direct pressure, nearby boreholes, or excavation history.
Question. Does the model simulate staged excavation or gradual unloading?	
Caveat	No. The computation starts from the already-open 2D geometry, with $\boldsymbol{u}(\boldsymbol{x}, 0) = \mathbf{0}$, no active initial stress tensor, and no active top load.
Question. What measured in-situ stress is findable before excavation?	
Gate	Cited CD-A papers give pre-excavation stress evidence, but not an active initial condition for the current run. The water-content modelling paper reports a measured in-situ stress state projected to the mesh with compressive total stresses $\sigma_h = 3\,\mathrm{MPa}$ and $\sigma_v = 7\,\mathrm{MPa}$ [30, Sec. 3]. The GETE modelling paper lists $\boldsymbol{\sigma}_0 = -(5, 5, 7) \times 10^6\,\mathrm{Pa}$ and notes that in-situ stress and pore pressure measurements are challenging and change with excavation [28, modelling Table 2]. The current active XML does not apply either value: `ic_sigma0` and `load_top` are defined but inactive, so use requires a component/sign convention, a principal-direction or rotation-angle decision, and an excavation-reference decision.
Question. Is a prestress angle or principal-stress orientation closed by the cited sources or XML?	
Gate	No. The cited papers give stress magnitudes and geological or structural orientation context, but not an accepted principal-stress rotation for the current 2D prestress tensor. CD-A bedding and niche-orientation evidence from the characterization source cannot be substituted for stress orientation without a modelling decision [29, Sec. 4.3]. A usable prestress definition would need principal directions or a rotation angle, component order, sign convention, value choice, and the excavation-reference state; the current active XML has no active initial-stress tensor.
Question. How should early pressure or displacement transients be interpreted?	
Gate	They may mix physical drainage with adjustment from the prescribed numerical initial state. Hard mechanical validation needs a prestress and excavation-reference decision first.

OGS Inputs

Question. Which OGS process and primary variables define the active model?	
Established	The active process is `THERMO_RICHARDS_MECHANICS` with primary variables `displacement`, `pressure`, and `temperature`. The finite-element orders are 2, 1, and 1, respectively.
Question. Which source hierarchy defines the equations and active implementation?	
Established	The active implementation is the transferred OGS project file and its included XML files. The equation notation follows the OGS TRM documentation and the Wang–Kosakowski–Kolditz TRM notation used in the report. If notation and checked XML differ, the XML defines the active CD-A run and the mismatch is recorded as a gate.
Question. Which numerical time-loop and solver settings are active?	
Checked input	The current run uses Backward Euler, fixed time stepping from 0 to $1.4 \times 10^8\,\mathrm{s}$, `basic_newton` with maximum 150 iterations, component-wise Δx convergence tolerances 10^{-10} absolute and 10^{-8} relative, and the `PardisoLU` linear solver.
Question. Which governing terms are explicitly inactive?	
Caveat	The current formulation has zero specific body force, no explicit vapor-transport phase terms, no active hydraulic source term, no active initial stress, and no active top load.

Question. Which parameter family is released in the first workflow?

Release policy Only the intrinsic-permeability tensor field supplied through `k_i_rd` is released as the first material field. The current modelling policy keeps porosity, retention, mechanics, thermal parameters, boundaries, and initialization fixed or gated.

Question. How do candidate-run parameter changes enter OGS?

Release policy They enter through current-run mesh fields or explicit XML/input replacements, such as `k_i_rd` in the candidate mesh. They are not silent edits to the governing equations or to the exchanged source-model provenance.

Question. How are permeability and porosity passed to OGS?

Established The current OGS project uses `MeshElement` fields: `k_i_rd` for intrinsic permeability and `n_rd` for porosity. The current `n_rd` field is treated as fixed at 0.105.

Question. Are retention and relative permeability fitted now?

Fixed context No. Saturation uses the fixed van Genuchten law with $S_{lr} = 0.1$, $S_{gr} = 0$, $m = 0.45$, and $p_b = 10\,\text{MPa}$. Relative permeability uses the matching fixed van Genuchten/Mualem-type law with minimum 10^{-25}.

Question. Are mechanical parameters fitted now?

Gate No. Orthotropic elasticity, $\alpha_B = 1$, Bishop exponent 1, and saturation-dependent swelling are active but fixed. Elastic-parameter release requires hydro-mechanical residuals plus prestress/excavation policy.

Question. Are thermal or liquid properties fitted now?

Caveat No. Density, viscosity, heat capacities, thermal conductivities, and thermal expansivity are fixed XML inputs. The `CTE`=1254.74 value remains a provenance caveat rather than a calibrated thermal-expansion value.

Question. Should any material parameter be time-dependent in this first pass?

Release policy No. Time-dependent $\boldsymbol{K}(\boldsymbol{x}, t)$, porosity, retention, fracture closure, and healing/sealing inputs are excluded until repeated same-support evidence separates material evolution from boundary forcing, saturation state, and spatial heterogeneity.

Model Outputs And Observation Operators

Question. Which VTU arrays are requested in the current OGS workflow?

Established The required output arrays are `pressure`, `saturation`, `temperature`, `displacement`, and `porosity`.

Question. Which process secondary variables are available but not required outputs here?

Checked input `velocity`, `sigma`, `epsilon`, and `swelling_stress` are process secondary variables. They can be requested or sampled only with an explicit output and interpretation policy.

Question. Is permeability an OGS state output?

Established No. $\boldsymbol{K}$ is an input mesh field, not a dynamic state variable. Permeability comparisons must sample or average the mesh field with the correct support and direction operator.

Question. How should OGS water content be formed from outputs?

Release policy The model-side mobile water-content proxy is $\theta = nS_l$, using porosity and liquid saturation. This is not automatically the same as total measured water content.

Question. How are output times tied to observations or boundary curves?

Gate The run starts at $t = 0$, identified by comments as 18 September 2019, and ends at 1.4×10^8 s. Measurements require an explicit date-to-model-time conversion and interpolation or matching rule.

Question. When does an OGS field become a residual?

Gate Only after an observation operator defines the measured quantity, support, time mapping, units, reference convention, and uncertainty. Otherwise the OGS field is only a diagnostic or model state.

Question. What does intrinsic permeability describe materially?	
Release policy	$\boldsymbol{K}(\boldsymbol{x})$ describes effective Darcy mobility of the clay/EDZ continuum before relative permeability and viscosity are applied. It is the base hydraulic material field in the current inversion stage.
Question. How does hydraulic conductivity relate to permeability?	
Release policy	Hydraulic conductivity is derived from intrinsic permeability, relative permeability, liquid density, viscosity, saturation state, and the chosen head/gravity convention. It is not an independent base field in the current OGS input.
Question. What does porosity describe and how does OGS use it?	
Fixed context	$n(\boldsymbol{x})$ is the effective pore-volume fraction used in storage, thermal mixing, and $\theta = nS_l$. It is represented by `n_rd`, but the current field is constant and not released.
Question. What do saturation and retention describe?	
Fixed context	S_l is the liquid occupancy of the OGS pore space. It is computed from $p_c = -p_l$ through the fixed retention law, and it controls storage, relative-permeability mobility, swelling, and water-content post-processing.
Question. What does relative permeability describe?	
Fixed context	$k_{\mathrm{rel}}(S_l)$ reduces liquid mobility under unsaturated conditions. It multiplies $\boldsymbol{K}/\mu_l$ in the Darcy velocity and is fixed by the current material law.
Question. Is storativity a separate released field?	
Gate	No. Storage behaviour follows from porosity, saturation derivative, liquid compressibility/density, and mechanical volumetric coupling. A separate storativity field would be a new modelling choice.
Question. How do Biot and Bishop fields enter the model?	
Fixed context	The fixed Biot coefficient is $\alpha_B = 1$, and the Bishop power law with exponent 1 gives $b(S_l) = S_l$. Together they scale the pore-pressure term in the mechanical balance.
Question. What is the elastic field description?	
Fixed context	The active mechanical model uses fixed orthotropic linear elasticity with $E = (13.8, 6.0, 13.8)$ GPa, $G = (4.79, 2.46, 4.79)$ GPa, and $\nu = (0.44, 0.22, 0.44)$.
Question. Does the same bedding angle control mechanics, permeability, and electrical anisotropy?	
Gate	The cited structural evidence supports bedding orientation as important context for permeability candidates, but it does not prove that Darcy, mechanical, and electrical anisotropy must share one hard angle in this chapter.
Question. What role do swelling fields have?	
Fixed context	The model includes a fixed saturation-dependent swelling law with 1 MPa swelling pressures in each direction and $0.1 \le S_l \le 1$ limits. It is active physics but not a released parameter family.
Question. How should faults, cracks, and EDZ features be classified here?	
Gate	They are not active discrete OGS geometry or separate material classes in this chapter. They can become structural priors, masks, equivalent fields, or scenarios only after their source geometry, scale, aperture meaning, and 3D-to-2D projection are defined.

1.5.1 Gate Map for Later Work

The gates are not additional questions outside the inventory. They are the subset of question rows in Table 1.13 whose answers are incomplete, implementation-dependent, or deliberately held fixed. Table 1.21 therefore records how each open gate should be read: first as the question that creates the gate, then as the modelling consequence if the gate remains unresolved. In this merged form, the chapter gives us a reproducible model baseline and a gate map at the same place where the questions are asked. Later chapters can discuss measurement packages and inversion decisions without redefining the PDE model or silently moving inactive XML parameters into the active physics.

Table 1.21: Relation between open gates and the question rows that create them.

Gate family	Question rows that create the gate	Consequence while unresolved
Pressure-boundary implementation	The hydraulic-boundary rows asking which pressure Dirichlet conditions are mathematically intended and which are actually attached in the checked XML.	The reference model may state pressure on all square sides, but the implementation must either show the left/right/bottom pressure support or document the preprocessing step that adds it.
Niche pressure-curve provenance	The hydraulic-boundary rows asking what is known and unknown about $p_E(t)$.	The numerical curve can be used as the active input, but source sensors, filtering, interpolation, time origin, and uncertainty remain unavailable for hard boundary-error modelling.
Initial pressure and excavation state	The initial-state rows asking whether the model encodes post-excavation drainage, staged excavation, prestress, and early transient interpretation.	Early pressure and displacement responses should not be used as hard HM evidence until the initial pressure field, prestress tensor, and excavation-reference policy are fixed.
Released OGS input fields	The OGS-input rows asking which parameter family is released and whether retention, mechanics, thermal/liquid properties, or time-dependent fields are fitted.	The first workflow releases only the intrinsic-permeability tensor magnitude field; other parameter families remain fixed context or staged-release gates.
Output-to-residual interpretation	The model-output rows asking how water content is formed, how output times map to observations, and when an OGS field becomes a residual.	OGS fields remain model states or diagnostics until an observation operator defines support, time mapping, units, reference convention, and uncertainty.
Derived material quantities	The material-field rows asking whether hydraulic conductivity, storativity, saturation/retention, relative permeability, Biot/Bishop coupling, and swelling are base fields or derived/fixed laws.	Derived quantities must stay labelled as derived; they should not be released or scored as independent material fields without a new modelling decision.
Structural features and missing evolution	The material-field row asking how faults, cracks, EDZ features, and healing/sealing should be classified.	These features may enter later as priors, masks, equivalent fields, or scenarios, but not as active discrete geometry or time-evolving material physics in this baseline chapter.

Chapter 2

Measurements and OGS Observation Links

2.1 Purpose and Evidence Boundary

This chapter records the measurement side of the CD-A Niche 4 modelling problem. The purpose is not to turn every available dataset into a likelihood term. The purpose is to state, stream by stream, what is measured, where and when it is measured, which unit and reference convention are used, and how the value can enter the OGS workflow as an input, output residual, material-field prior, diagnostic, or caveat.

The primary evidence for the plotted data and row summaries is the set of email- and TeamBeam-provided source files named in the bibliography. We use the processed observation tables and generated measurement figures copied into this paper as traceable reductions of those files. The processed-observation layer is the authority for row counts, time ranges, coordinate status, and current activation gates [5, source filenames and transfer context]. Public internet sources were checked only where the provided source catalogue lacks residual-ready numeric exports. They confirm the broader CD/CD-A monitoring context, including pressure, extensometer, crackmeter, Taupe/TDR, RH, water-content, and resistivity streams, but they do not replace the missing project-specific Geoscope, pressure, or crack-opening time series [26, abstract and Sec. 2] [10, Secs. 2.4 and 3.2].

The geometry rule is the same for all streams. A measurement is not an OGS datum until its support is mapped into the current two-dimensional model frame. Point measurements need coordinates and a sampling rule. Borehole interval measurements need segment geometry and interval weights. ERT needs a mesh-to-mesh observation operator. Taupe/TDR needs line-band support. HM monitoring needs a reference frame, instrument sign convention, and a 3D-to-2D reduction policy before it can be weighted.

Table 2.1: Measurement stream roles in the current OGS workflow.

Stream	Current status	OGS tie	Main gate or use
Permeability pulse tests	Active material likelihood	Direct interval-scale observation of intrinsic permeability $\boldsymbol{K}(\boldsymbol{x})$.	Usable for 75 mapped open-twin BCD-A32/A33 rows; support conflicts and older endpoint geometry remain gates.
NMR water content	Active but provisional state likelihood	Observation of water-content state, compared to $\theta = nS_l$ after an observation policy.	Raw absolute water content is caveated by bound/interlayer water; trend/anomaly is the preferred next residual.
ERT	Diagnostic state field	Forward operator from S_l to water content to bulk resistivity to projected ERT resistivity.	Needs accepted coordinate transform, support mask, and covariance before likelihood use.

Table 2.1: Measurement stream roles in the current OGS workflow (continued).

Stream	Current status	OGS tie	Main gate or use
Taupe/TDR	Diagnostic trend field	EDZ-band trend operator tied to $\theta = nS_l$, or to dielectric/water-content calibration if confirmed.	Absolute unit and calibration are not confirmed; A3/A4 are mapped, A7/A8 are outside current support.
RH/suction	Boundary-condition check	Kelvin conversion to pressure or suction candidate for the open-niche boundary $p_E(t)$.	Active curve provenance, filtering, time origin, and uncertainty remain open.
Direct pressure sensors	Blocked future pressure residual	Candidate samples of $p_l(\boldsymbol{x}, t)$ or hydraulic head.	Mini-piezometer numeric time series, units, reference convention, and quality flags are missing.
Fault/fracture geometry	Structural prior or scenario field	Candidate prior for permeability, porosity, EDZ masks, or 2D projection scenarios.	3D geometry, measured aperture/effect, timing, and projection policy are incomplete for hard use.
Other HM monitoring	Blocked or qualitative HM validation	Candidate displacement, strain, crack opening, levelling, scan-difference, or pressure residuals.	Layout and levelling summaries exist; hard residuals need numeric exports, timestamps, units, reference zeros, and uncertainties.

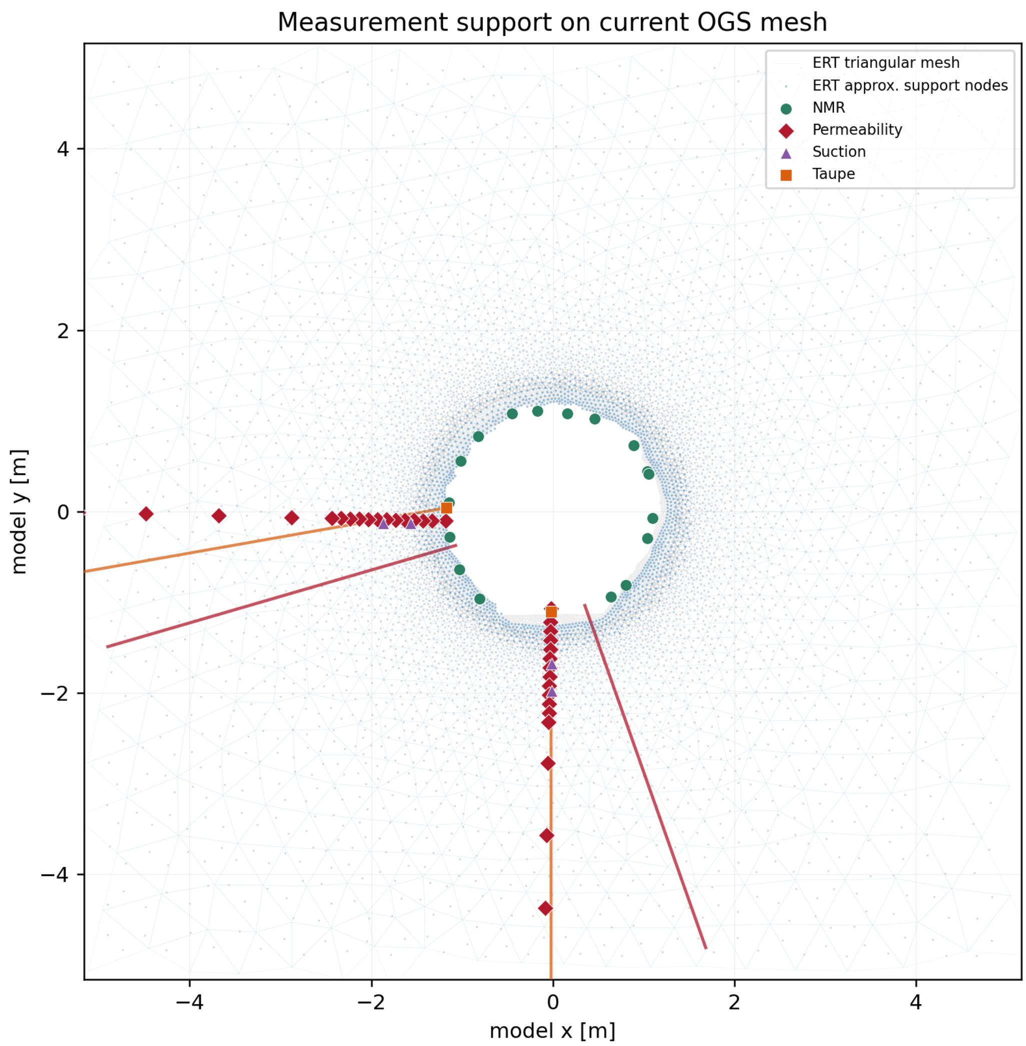

Figure 2.1: Mesh-mapped measurement supports in the current OGS domain. The plot separates projected ERT cells, NMR labels, Taupe/TDR supports, RH/suction points, and permeability pulse-test intervals. Streams without accepted OGS-frame support are handled separately in their own sections.

2.2 Direct Permeability Pulse Tests

Meaning. The permeability stream records interpreted pulse-test permeability values, transmissibility values, and raw pressure-decay traces. The source method is a modified double-piston packer test with a nominal 10 cm interval and nitrogen injection. Therefore the model-facing quantity is a gas-test interpretation of intrinsic permeability, not hydraulic conductivity, not liquid relative permeability, and not a direct tensor component. Gas apparent permeability has its own pressure dependence, so the liquid-equivalent intrinsic value must remain separated from the OGS relative-permeability factor [12, abstract] [15, abstract and Secs. 2.2, 3.2].

Position in the domain. The current active supports are the mapped BCD-A32 and BCD-A33 open-twin intervals inside the Niche 4 OGS mesh. Figure 2.2 shows these supports as borehole-line and cell-sampling locations. Older rows from BCD-A24, BCD-A25, BCD-A26, BCD-A27, and BFM-D19 remain visible in the catalogue, but they do not have accepted endpoint geometry or current-domain support for the first direct fit.

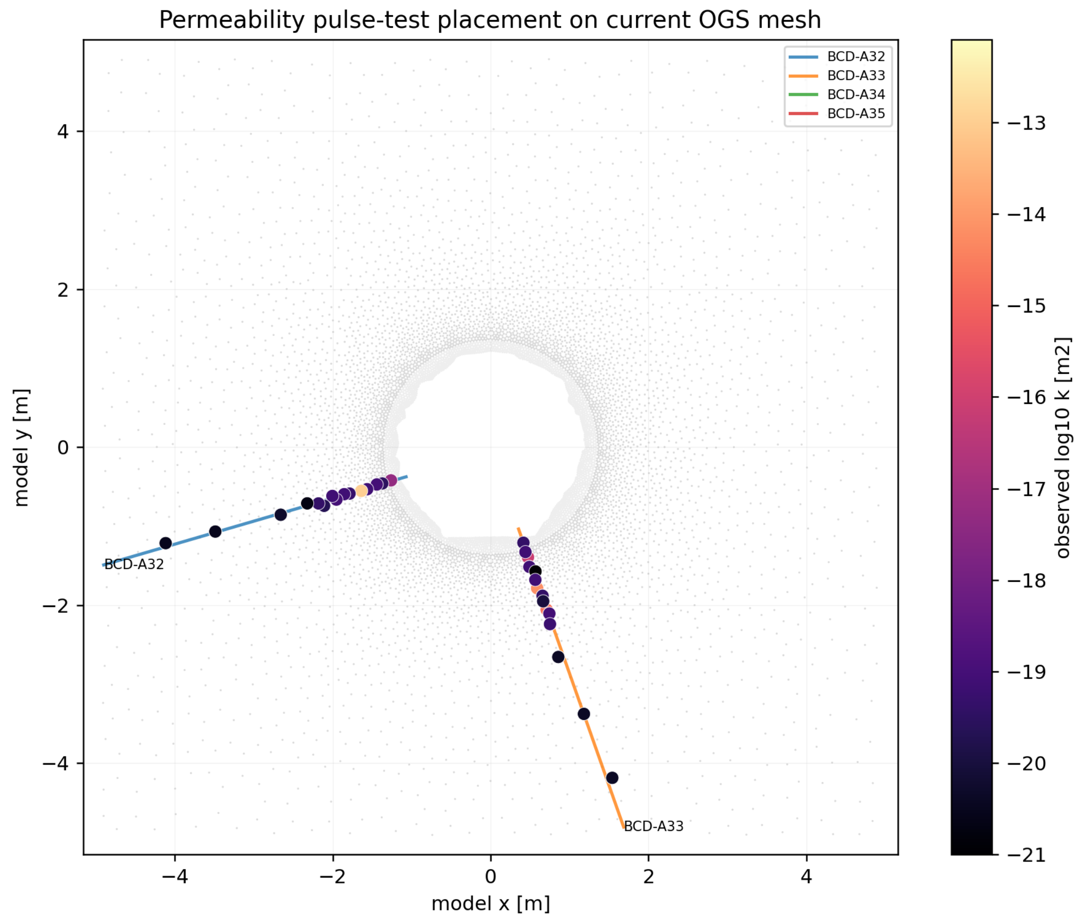

Figure 2.2: Location of active direct permeability supports on the current OGS mesh. The plotted supports are the BCD-A32 and BCD-A33 pulse-test intervals currently usable as direct constraints on $\boldsymbol{K}(\boldsymbol{x})$.

Time of measurement. The processed table contains 204 interpreted rows and 6687 pressure-decay rows. The current objective uses the 2024 BCD-A32/A33 rows as quasi-static material evidence. The 2021 rows are retained as open/closed and historical context, but they are not part of the current active objective unless their support policy is reopened.

Measured value, units, and OGS tie. The interpreted value is permeability in m^2, used in log10 space. The active usable rows span roughly $-21.0 \leq \log_{10}(k/\mathrm{m}^2) \leq -12.10$. The OGS tie is direct: each row constrains an interval-weighted directional response of the intrinsic permeability tensor,

$$r_i^k := \log_{10} k_{i,\mathrm{pred}} - \log_{10} k_{i,\mathrm{obs}}, \qquad k_{i,\mathrm{pred}} = \sum_{c\in\mathcal{C}_i} w_{ic}\, \mathbf{e}_i^\top \boldsymbol{K}_c \mathbf{e}_i.$$

Here $\mathcal{C}_i$ is the set of OGS cells sampled by row i, w_{ic} are support weights, $\mathbf{e}_i$ is the interval or borehole-axis unit direction, and $\boldsymbol{K}_c$ is the cell-wise intrinsic-permeability tensor. This residual acts on the material field used by Darcy's law, so permeability changes also affect pressure diffusion, saturation evolution, NMR water-content predictions, ERT resistivity predictions, and Taupe/TDR trends.

Uncertainty and reliability. The first-pass direct residual uses a 0.5 log10-unit standard deviation and duplicate-aware weights. This is a policy choice consistent with the order of the experimental/evaluation uncertainty, not a proof that every support conflict is resolved. Same-support rows can differ by several log10 units, so outlier policy, support aggregation, and endpoint recovery remain active gates.

Current computational use.

With the available source data now, the defensible use is a direct material-field likelihood on the mapped positive BCD-A32/A33 rows. The nitrogen pulse-test value is treated as a noisy scalar observation of the liquid-equivalent intrinsic permeability used by Darcy flow. The observation operator is the interval average of $\mathbf{e}^\top\boldsymbol{K}\mathbf{e}$ over the mapped support, evaluated in log10 space. This is the best current transfer from a gas pulse test to OGS $\boldsymbol{K}$: keep the reported value as intrinsic permeability in m^2, keep the gas/slip interpretation caveat visible, and absorb unresolved gas-to-liquid and support interpretation errors through a conservative log-space uncertainty and duplicate-aware weighting [12, abstract] [15, abstract and Secs. 2.2, 3.2] [5, source files used for the generated support and row summaries].

This stream directly constrains only $\boldsymbol{K}(\boldsymbol{x})$ or its released parameterization. It does not independently constrain pressure, saturation, porosity, retention, or relative permeability. If the unresolved gas correction, interval support, tensor direction, or duplicate policy is wrong, an inverse run will express that error as false spatial heterogeneity in $\boldsymbol{K}$.

Grounding questions.

Table 2.2: Grounding questions for direct permeability pulse tests.

Question. How should the nitrogen pulse test be linked to OGS $\boldsymbol{K}$?	
Ready	As an interval-scale observation of liquid-equivalent intrinsic permeability $\mathbf{e}^\top\boldsymbol{K}\mathbf{e}$, in m^2, not as hydraulic conductivity, relative permeability, or saturation.
Question. Was a Klinkenberg or gas-slip correction already applied?	
Gate	Not proven by the available source files; the provider must state the correction, gas pressure range, temperature, leakage screening, and fitting method for each interpreted row.
Question. Is the 10 cm interval operator accepted for inversion?	
Gate	Provisionally yes for BCD-A32/A33; final use needs accepted endpoints, cell weights, and a policy for older BCD-A24/A25/A26/A27/BFM-D19 rows.
Question. How should repeated or conflicting supports be weighted?	
Policy	Use duplicate-aware or robust weights until the collaborator confirms whether repeated rows are independent tests, reprocessed duplicates, or evidence of real time evolution.
Question. What would fully ground this likelihood?	
Gate	A row-level table with endpoints, direction, date, gas interpretation metadata, quality flags, support group id, and accepted log-space uncertainty.

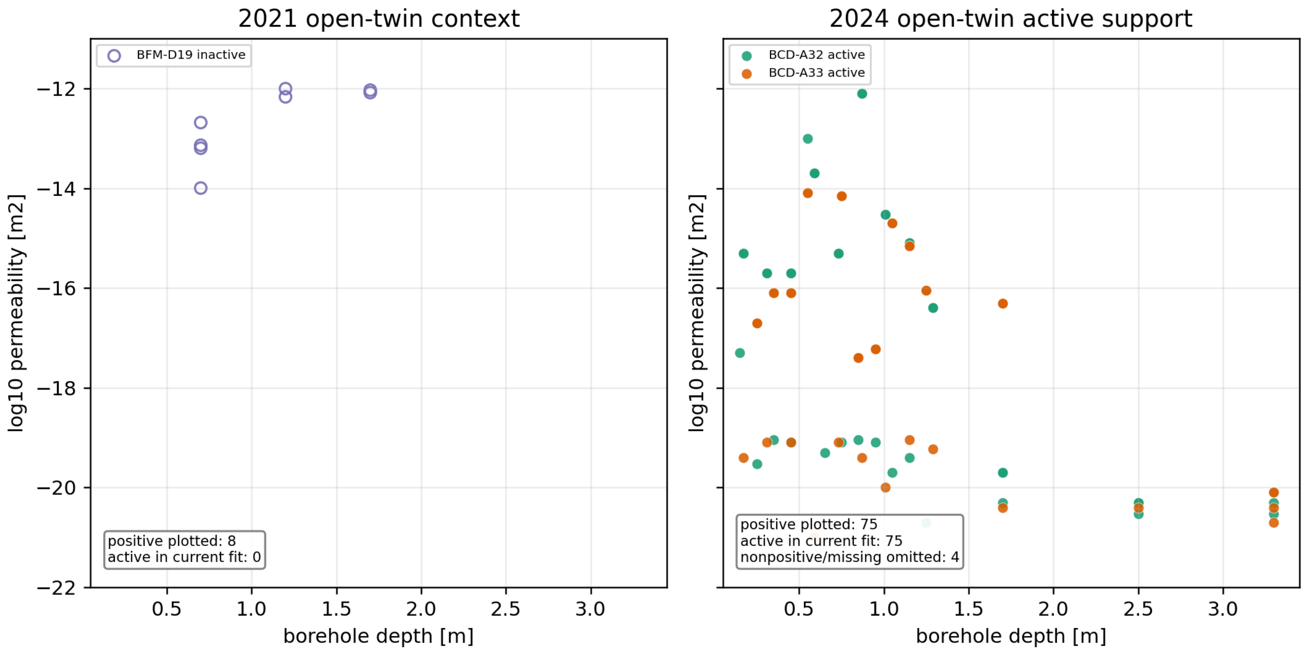

Figure 2.3: Open-twin interpreted permeability rows by campaign year. The 2024 BCD-A32/A33 values form the active support; earlier rows remain context until geometry and support policy are accepted.

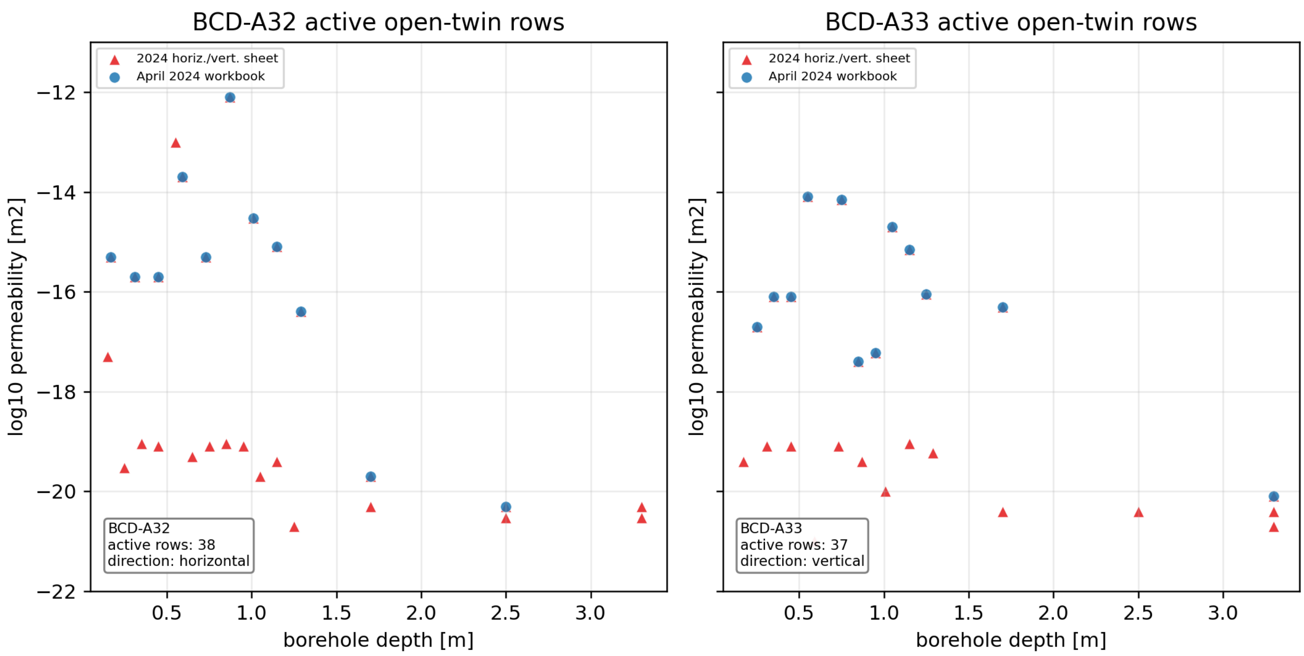

Figure 2.4: Active open-twin permeability rows by borehole. BCD-A32 and BCD-A33 provide the current horizontal and vertical support families for the direct material-field likelihood.

2.3 NMR Water-Content Measurements

Meaning. NMR is the strongest provided water-content evidence, but it is not identical to OGS liquid saturation. The weekly and seasonal source tables report volumetric water content in vol.% and relaxation-time information. In claystone the NMR signal can include mobile pore water and bound or interlayer water. The model-side quantity $\theta = nS_l$ is therefore only the mobile-water proxy unless a bound-water observation model is added [30, Sec. 4.2] [9, Secs. 1 and 2.5] [8, single-sided NMR technical slides].

Position in the domain. The processed NMR labels are mapped into the two-dimensional OGS frame when coordinates exist. Figure 2.5 shows the active and contextual support positions. Niche 4 support is relevant for the current model. Niche 3 campaign rows are retained as experimental context, not as direct observations of the present slice.

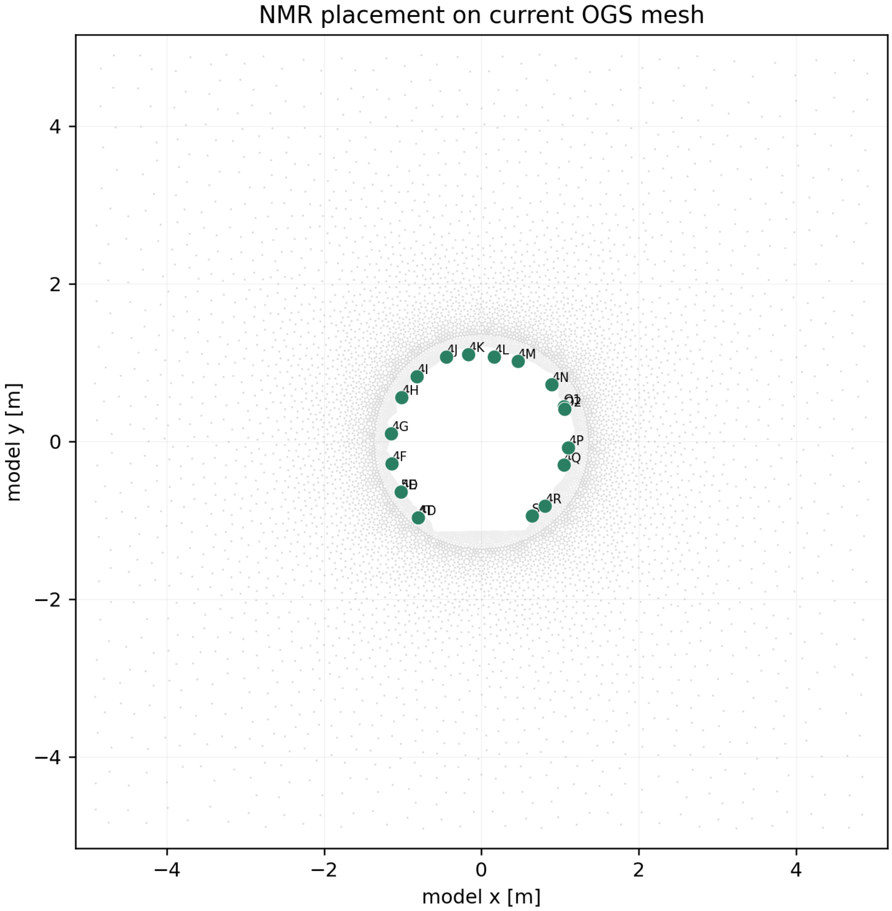

Figure 2.5: NMR support positions in the OGS model frame. These labels define where $\theta = nS_l$ is sampled from OGS outputs for raw and trend/anomaly comparisons.

Time of measurement. The processed NMR layer contains 170 weekly rows and 294 seasonal profile rows. The weekly 4E/4S monitoring covers 2021-10-06 to 2025-09-09, while the seasonal archive contains campaign profiles for repeated Niche 3 and Niche 4 measurement dates. The row-level table keeps caveats such as the 2025 4E detuning period.

Measured value, units, and OGS tie. The measured water content is reported in vol.%; in the model we use the fraction $\theta_{\text{obs}} = \text{vol.\%}/100$. The natural OGS comparison is

$$\theta_{\text{NMR}}^{\text{obs}} = \theta_{\text{mobile}} + \theta_{\text{bound}} + \epsilon_{\text{NMR}}, \qquad \theta_{\text{mobile}}^{\text{model}} \approx nS_l.$$

Here θ_{bound} represents NMR-visible bound or interlayer water not carried as an OGS mobile-liquid state, and ϵ_{NMR} represents measurement and observation-model error. The active objective uses raw absolute θ

on the accepted mapped subset. The preferred next policy is a within-label anomaly residual, because it removes constant station or bound-water offsets before comparing model and measurement changes.

Uncertainty and reliability. The bound-water check shows why absolute NMR must remain provisional: with fixed $n = 0.105$, 315 of 464 NMR target rows exceed the maximum possible value of nS_l if interpreted as mobile water. In the usable current-mesh subset, 162 of 287 rows exceed the same bound. Reported confidence intervals and an uncertainty floor are useful, but they do not solve the semantic difference between total NMR-visible water and mobile OGS liquid.

Current computational use.

With the available source data now, NMR can be used as a provisional water-content-state comparison, preferably as a within-label trend/anomaly likelihood rather than a raw absolute likelihood. The current OGS state supplies $\theta_{\text{model}} = nS_l$. The NMR value should be treated as total NMR-visible water unless the source processing proves a mobile/free-water cut. The safest current observation model is therefore

$$\theta_{\text{NMR}}^{\text{obs}} = nS_l + b_{\text{label}} + b_{\text{campaign}} + \epsilon,$$

or the equivalent anomaly model after subtracting stable label/campaign offsets. This matches the provided NMR evidence that many rows exceed the fixed OGS porosity and the cited clay/NMR caveat that NMR-visible water can include bound or interlayer water [30, Sec. 4.2] [9, Secs. 1 and 2.5] [5, NMR source files used for the generated row summaries].

The inverse role is to constrain pressure/saturation evolution after the boundary and permeability field drive S_l. NMR alone cannot separate porosity, retention, mobile fraction, and bound-water offset. Releasing those at the same time without priors would let the inverse problem fit the same water-content residual in several physically different ways.

Grounding questions.

Table 2.3: Grounding questions for NMR water-content measurements.

Question. What exactly does the supplied NMR value measure?	
Gate	Treat it as total NMR-visible volumetric water content for now; provider confirmation is needed for any mobile/free-water interpretation.
Question. How should NMR be linked to OGS output now?	
Policy	Compare to $\theta = nS_l$ only through a bias or anomaly observation model, not as raw mobile water everywhere.
Question. Can bound/interlayer water be anchored simply?	
Gate	Yes, if collaborators confirm that the bound-water offset is approximately constant by label, material class, depth, or campaign; then it can be a nuisance offset rather than an OGS state.
Question. Could bound water be post-processed from OGS saturation?	
Gate	Only if a supported relation is supplied; otherwise it should remain a constant or hierarchical observation offset, not a fitted material field.
Question. What would fully ground NMR residuals?	
Gate	Source units, T2/mobile-water cut policy, calibration and detuning flags, label/campaign offsets, support coordinates, confidence-interval conversion, and a rule for rows where $\theta_{\text{obs}} > n$.

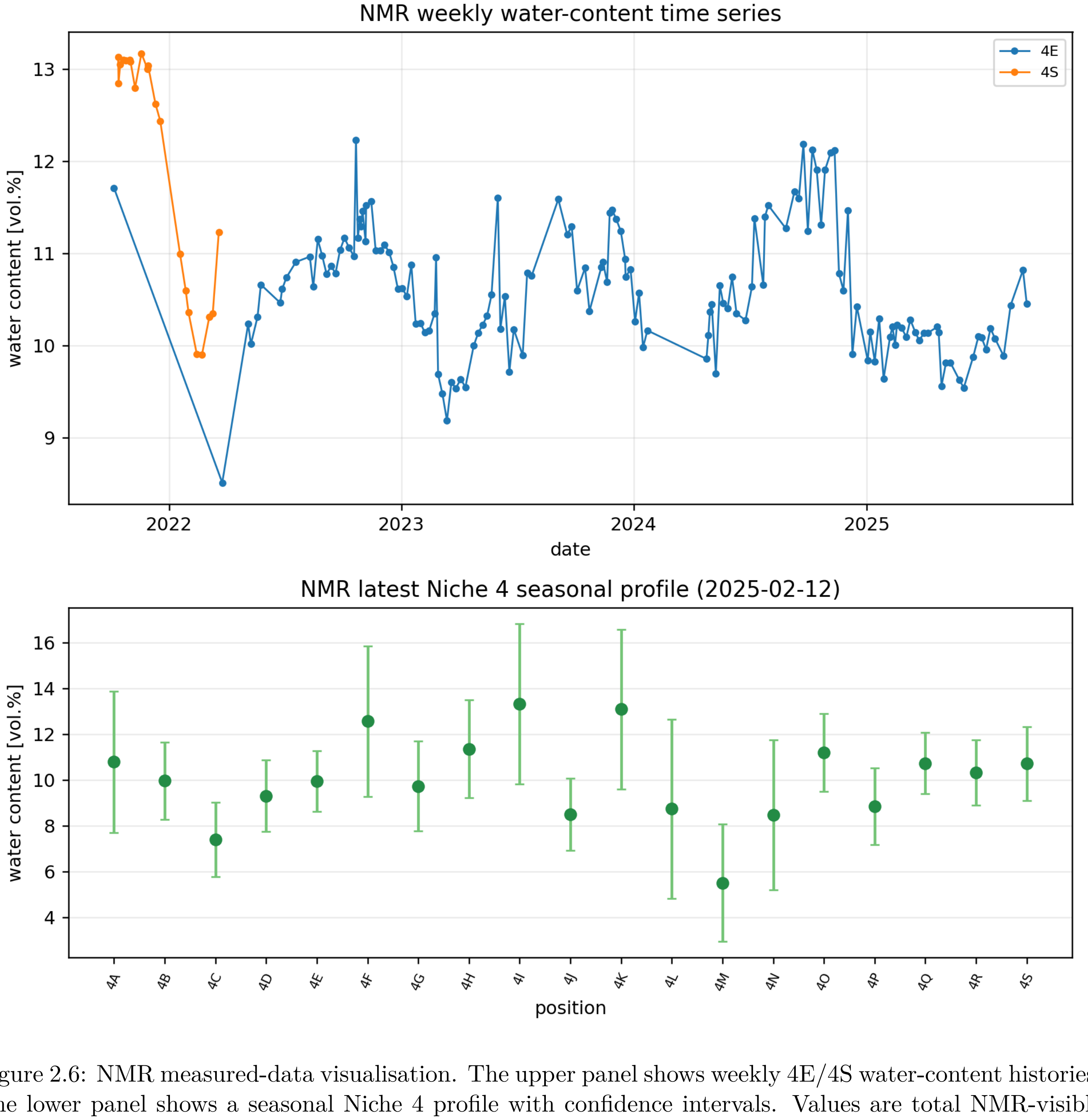

Figure 2.6: NMR measured-data visualisation. The upper panel shows weekly 4E/4S water-content histories. The lower panel shows a seasonal Niche 4 profile with confidence intervals. Values are total NMR-visible water content in vol.%.

2.4 ERT Resistivity Measurements

Meaning. ERT measures an electrical response that is inverted to a resistivity field. It is not a direct measurement of saturation, pressure, or permeability. The physically defensible OGS tie is a forward observation operator: OGS predicts pressure and saturation; saturation is converted to a water-content proxy; water content is converted through the current empirical resistivity relation; the result is projected to the ERT support. Classical Archie-type relations motivate the structure, but claystone requires CD-A-specific calibration because surface conductivity, bound water, fractures, and anisotropy can contribute to electrical conduction [2, pp. 54–56] [20, abstract and Secs. 2–4] [30, Secs. 4.1–4.3].

Position in the domain. The ERT archive contains a reference inverted mesh with 5966 triangular cells and 3126 points. The provisional projection transforms raw ERT coordinates by $x_{\text{model}} = x_{\text{ERT}}$ and $y_{\text{model}} = z_{\text{ERT}} + 500\,\text{m}$. It maps 4676 cells inside an OGS cell, with 2035 cells both inside an OGS cell and inside the approximate near-niche support.

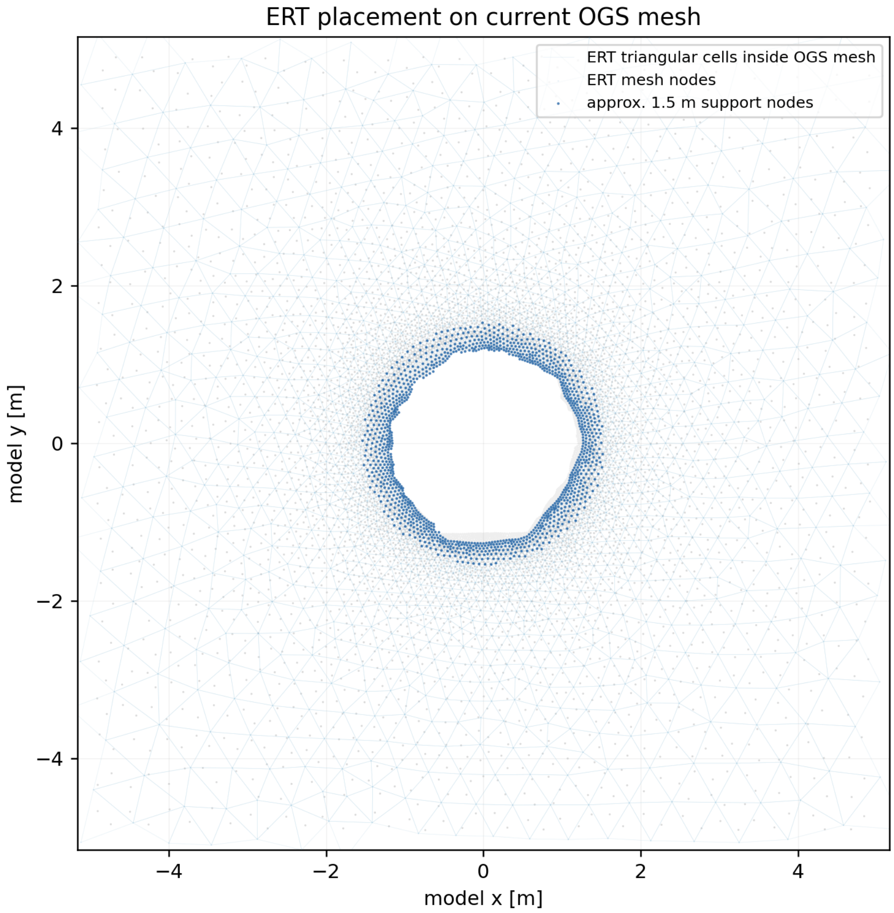

Figure 2.7: Projected ERT mesh support in the OGS frame. The darker subset indicates the current approximate near-niche support used for diagnostics before the exact support mask is accepted.

Time of measurement. The processed ERT inventory contains 1691 timestep entries from 2019-11-01 to 2024-12-31. Of these entries, 1675 have matching VTK fields and 16 are missing a matching VTK member. The archive therefore has dense temporal information, but it must be matched to OGS output times through an explicit tolerance.

Measured value, units, and OGS tie. The comparison should be made in log-resistivity, for example

$$S_l(\boldsymbol{x},t) \longrightarrow \theta(\boldsymbol{x},t) = nS_l(\boldsymbol{x},t) \longrightarrow \rho_{\Omega\text{m}}(\boldsymbol{x},t) = 1.108\,\theta(\boldsymbol{x},t)^{-1.58} \longrightarrow \mathcal{O}_{\text{ERT}}[\rho_{\Omega\text{m}}],$$

followed by

$$r^{\text{ERT}} := \log_{10}\rho_{\text{pred}} - \log_{10}\rho_{\text{ERT}}.$$

Here $\rho_{\Omega\mathrm{m}}$ denotes electrical resistivity in $\Omega\,\mathrm{m}$, and $\mathcal{O}_{\mathrm{ERT}}$ denotes the projection or aggregation from the OGS field to the inverted ERT support. The empirical relation is the current provisional default over the calibrated range $0.01 \leq \theta \leq 0.18$. It should not be interpreted as a universal Opalinus-Clay water-content law.

Uncertainty and reliability. ERT remains diagnostic. The unresolved gates are the exact coordinate transform, the exact tunnel/niche support mask, the 35 cm rock-depth convention, and the uncertainty/correlation model. Without a covariance or aggregation rule, dense neighbouring inversion cells would overweight one physical field image.

Current computational use.

With the available source data now, ERT should remain a diagnostic field and candidate future likelihood. The best current operator is the explicit post-processing chain

$$S_l \to \theta = nS_l \to \rho_{\Omega\mathrm{m}} = 1.108\,\theta^{-1.58} \to \mathcal{O}_{\mathrm{ERT}}\rho \to \log_{10}\rho,$$

using the provisional coordinate transform and near-niche mask only for diagnostics. This keeps the empirical CD-A water–resistivity relation separate from the OGS hydraulic state, and it prevents the inverted ERT mesh from being mistaken for independent point measurements [2, pp. 54–56] [20, abstract and Secs. 2–4] [30, Secs. 4.1–4.3] [5, ERT source files used for the generated field summaries].

For inversion, ERT mainly constrains the saturation/water-content pattern and therefore tests permeability, boundary forcing, and retention indirectly. Its dense cells share electrodes, inversion regularization, and projection errors. Without aggregation or covariance, ERT would dominate the objective numerically while adding much less independent information than the row count suggests.

Grounding questions.

Table 2.4: Grounding questions for ERT resistivity measurements.

Question. What ERT value should be compared?	
Diagnostic	Use log10 resistivity as the first defensible comparison; relative change or water-content products need separate source confirmation.
Question. How does ERT tie to OGS now?	
Diagnostic	Through $\theta = nS_l$ and the current empirical resistivity transform, followed by a mesh/support projection.
Question. Does bound water conduct like free water?	
Gate	Not settled; clay surface conduction and bound/interlayer water are explicit gates before absolute water-content inversion.
Question. Does electrical anisotropy matter?	
Gate	Possibly; bedding, fractures, and permeability anisotropy may affect electrical response, but no accepted electrical-anisotropy operator exists for this workflow.
Question. What would fully ground ERT likelihood use?	
Gate	Accepted coordinate transform, VTK field meaning, near-niche mask, reference date, bad timestep flags, bad electrode flags, aggregation or covariance, and a confirmed resistivity–water relation with uncertainty.

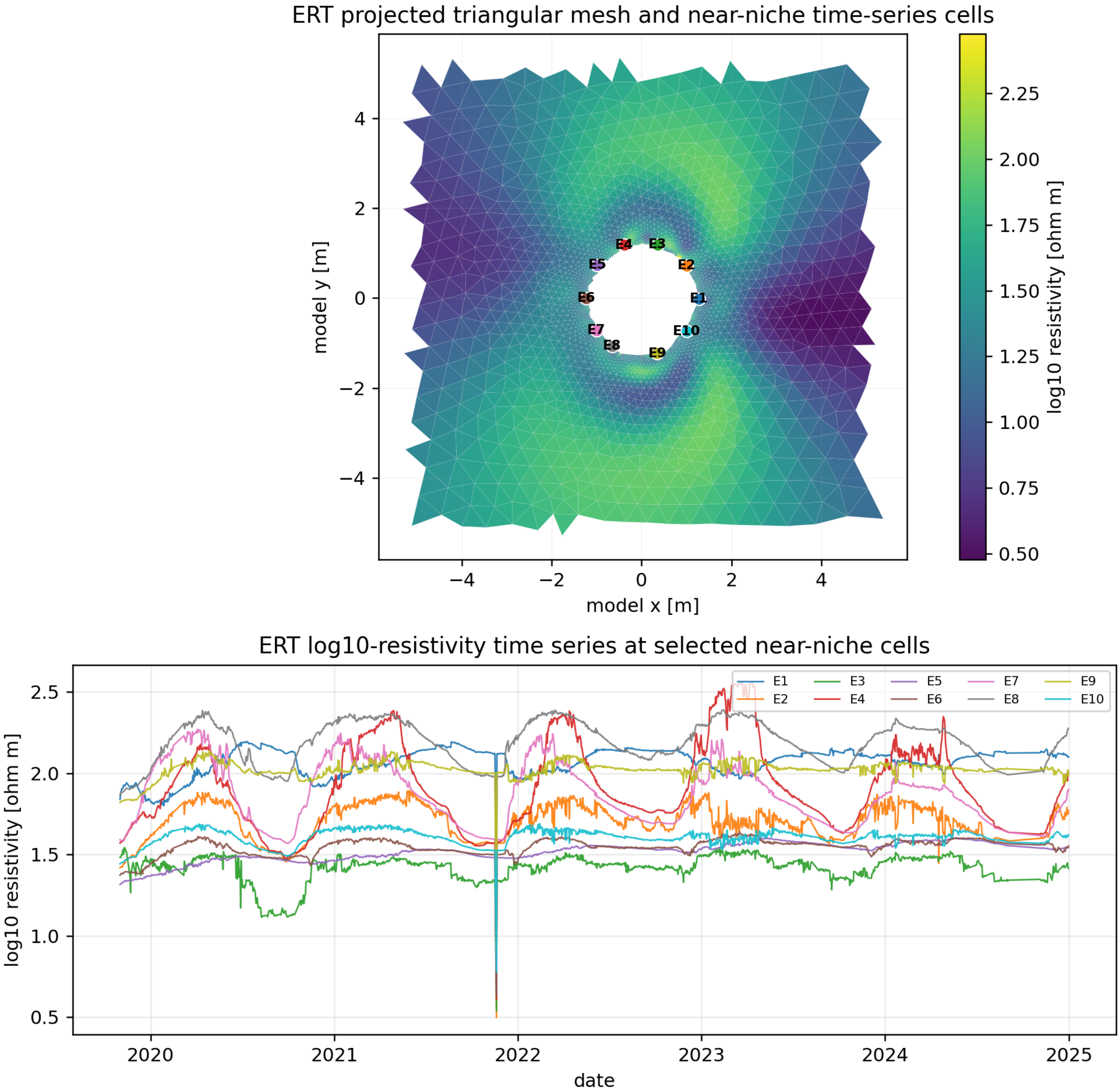

Figure 2.8: ERT measured-data visualisation. The upper panel shows a projected reference log10-resistivity field and selected support cells; the lower panel shows their log10-resistivity histories over matched ERT timesteps.

2.5 Taupe/TDR Measurements

Meaning. The Taupe/TDR workbook provides a dielectric or ARDP-derived water-content proxy. The provided files do not yet prove whether the values are calibrated volumetric water content, apparent relative dielectric permittivity, or another Taupe-specific proxy. For that reason the safe current interpretation is a same-series trend diagnostic, not an absolute saturation residual. General TDR literature supports the link between dielectric response and water content, but the CD-A claystone calibration must be supplied before absolute use [29, Sec. 4.4] [21, review and Secs. 1–2] [13, Taupe/TDR technical slides].

Position in the domain. The source sheets A3, A4, A7, and A8 correspond to Taupe supports and EDZ distance bands. A3 and A4 are mapped into the current OGS frame. A7 and A8 are kept in the workbook and operator tables, but their support falls outside the present model mesh.

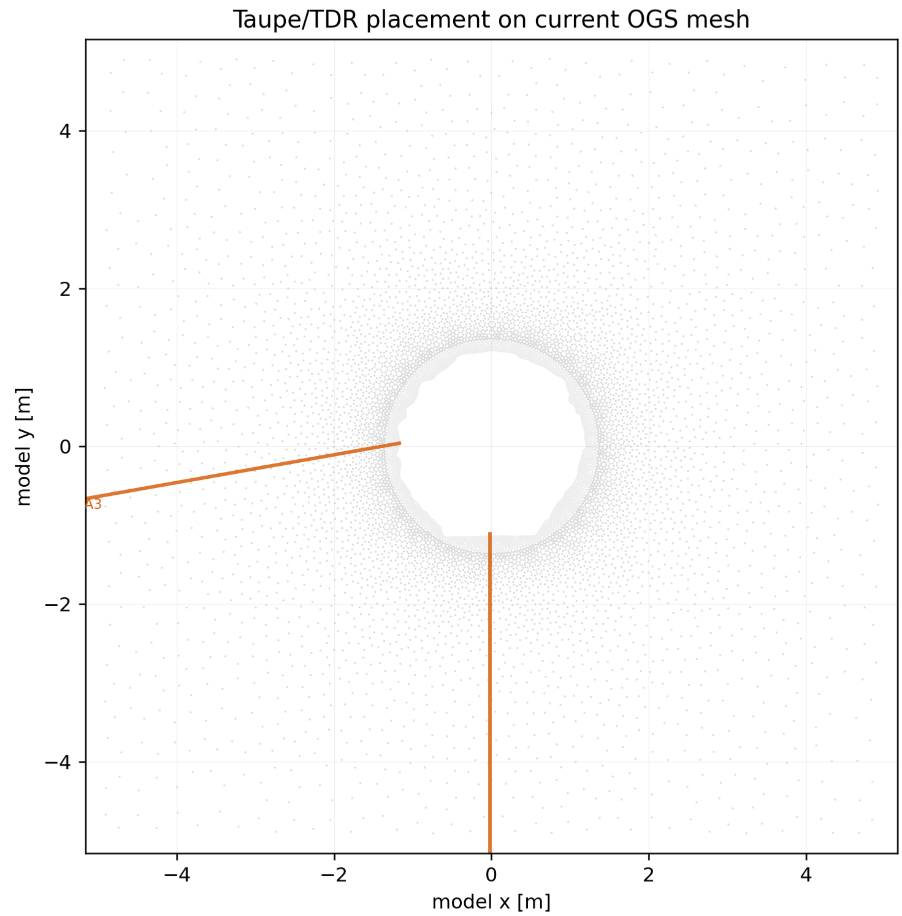

Figure 2.9: Mapped Taupe/TDR line supports. A3 and A4 are the current in-domain supports for trend diagnostics; A7 and A8 remain source-data context.

Time of measurement. The processed Taupe/TDR layer has 5088 sensor-date-band rows from 2019-12-01 to 2025-10-10. The rows are grouped by sensor and EDZ band, including 0–10, 10–20, 20–30, 30–40, 40–50, and aggregate 0–50 cm bands from the niche wall.

Measured value, units, and OGS tie. The workbook values span approximately 9.87 to 11.00 in source units. Until the unit is confirmed, the model comparison is the standardized trend

$$r_{\text{Taupe}} := \widehat{\Delta\theta_{\text{model}}} - \widehat{\Delta T_{\text{Taupe}}}, \qquad \theta_{\text{model}} = nS_l,$$

where hats denote a within-series normalization by sensor and EDZ band, and T_{Taupe} denotes the source Taupe/TDR series value rather than OGS temperature T. If a calibration is supplied, the same support can become an absolute water-content or dielectric residual.

Uncertainty and reliability. The trend operator is ready as a diagnostic, but absolute residual weights are not. The missing items are the workbook unit, calibration equations, baseline convention, sensor and band uncertainties, and whether bound/interlayer water affects the dielectric response in the same way as mobile pore water.

Current computational use.

With the available source data now, Taupe/TDR can be used as a same-sensor, same-band trend diagnostic for mapped A3/A4 supports. The current operator should band-average $\theta_{\text{model}} = nS_l$ over the corresponding EDZ-depth support, subtract a series baseline, standardize within sensor and band, and compare only the timing, sign, and relative amplitude of changes. This avoids pretending that the workbook unit is already an absolute volumetric water content or dielectric permittivity. The literature supports the general TDR water-content link, while the Taupe stream still needs CD-A-specific calibration before absolute use [21, review and Secs. 1–2] [13, Taupe/TDR technical slides] [5, Taupe source workbook and support files].

The inverse role is indirect: Taupe/TDR can help reject permeability or boundary candidates that produce the wrong wetting/drying trend near the EDZ. It cannot yet identify absolute saturation, porosity, retention, or bound-water response because the sensor unit and calibration are still open.

Grounding questions.

Table 2.5: Grounding questions for Taupe/TDR measurements.

Question. What does the Taupe workbook value represent?	
Gate	Unknown from the available source files; treat it as a source-unit dielectric/water-content proxy until calibrated units are confirmed.
Question. How should Taupe/TDR be linked to OGS now?	
Diagnostic	Compare standardized EDZ-band trends against band-averaged $\theta = nS_l$ changes for A3/A4.
Question. Can it become an absolute residual?	
Gate	Yes, if calibration equations map the workbook value to volumetric water content, apparent relative permittivity, or another modelled quantity with uncertainty.
Question. How should bands and sensors be weighted?	
Policy	As correlated groups by sensor, cable, band, and time until covariance or independent-noise assumptions are supplied.
Question. What would fully ground Taupe/TDR use?	
Gate	Unit definition, calibration constants, baseline date, support geometry by band, temperature, salinity, or conductivity corrections if used, quality flags, and sensor-band covariance.

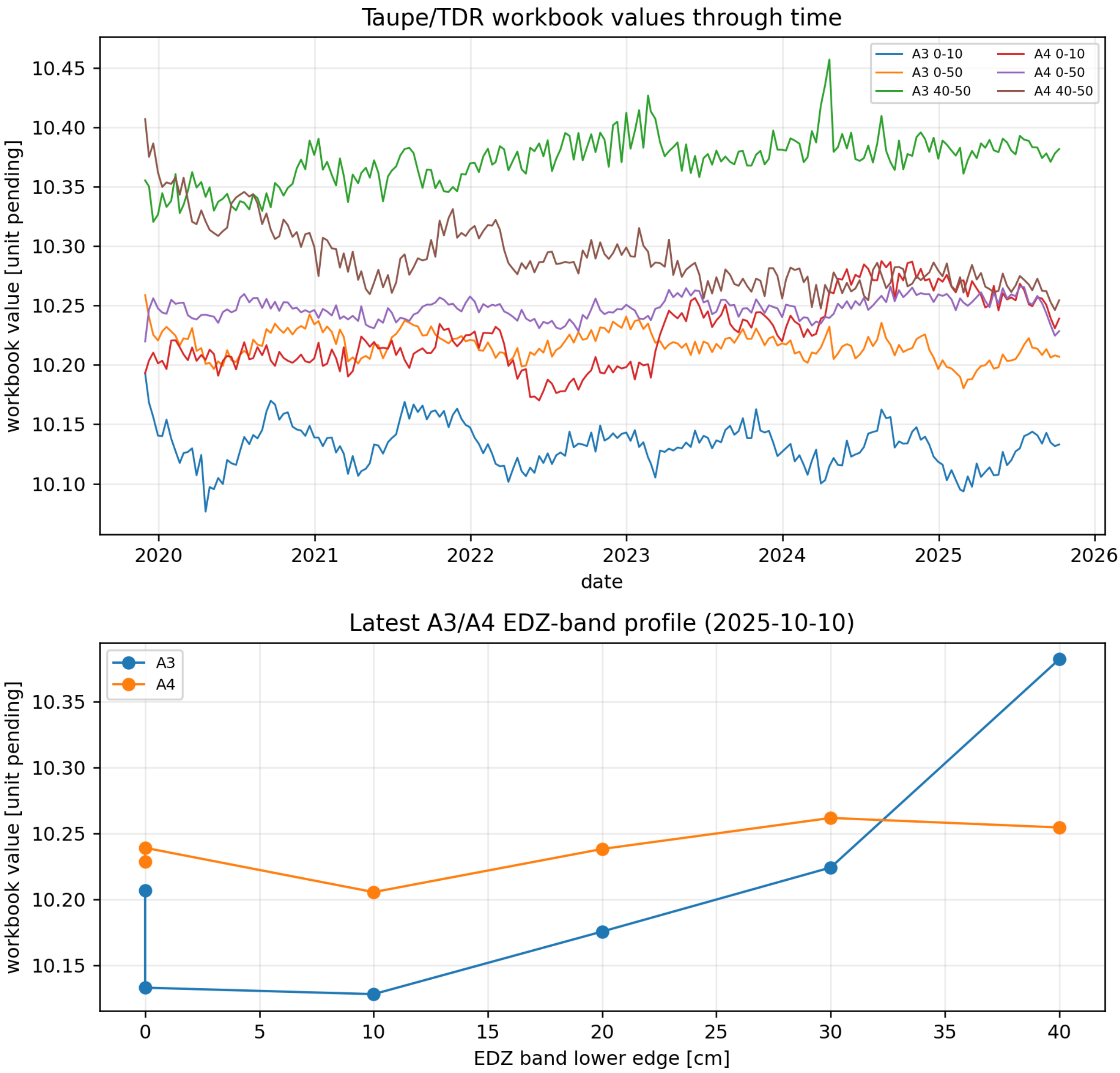

Figure 2.10: Taupe/TDR measured-data visualisation. The upper panel shows representative A3/A4 EDZ-band histories; the lower panel shows a latest EDZ-band profile. Values are retained in source workbook units.

2.6 RH, Suction, and Open-Niche Pressure Forcing

Meaning. The RH workbooks measure relative humidity, not liquid pressure directly. Under a Kelvin-equation convention, RH can be converted to a capillary-pressure or gauge-liquid-pressure candidate. This makes RH primarily a boundary-condition source or check for the open niche. It can later inform retention or boundary uncertainty, but it is not an interior material residual.

Position in the domain. The RH/suction supports are near the open-niche boundary and are mapped as boundary-check points in the OGS frame. Their role is to test whether the time-dependent boundary curve $p_E(t)$ is consistent with humidity-derived suction evidence.

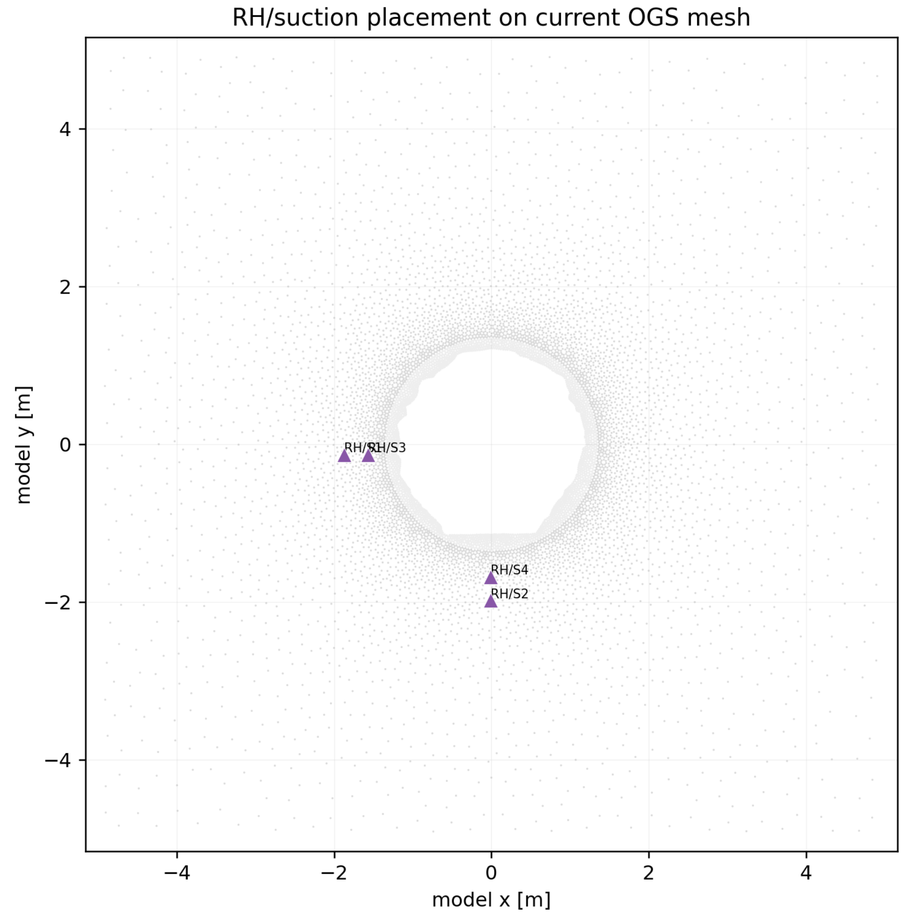

Figure 2.11: Mapped RH/suction support points used to check the open-niche pressure boundary. These are boundary-forcing supports, not cell-wise material observations.

Time of measurement. The processed RH layer contains 4247 daily rows from 2021-12-16 to 2025-09-04 for RH5, RH6, RH7, and RH8. RH5 and RH6 are the cleanest open-twin evidence. RH7 and RH8 contain low outliers near 13% RH and require stronger screening.

Measured value, units, and OGS tie. The source unit is RH in percent. We convert it to a gauge-pressure candidate by

$$p_l^{\text{Kelvin}} = \frac{\rho_l RT}{M_w} \ln(\text{RH}),$$

where RH is a fraction, $T = 298.15\,\text{K}$, and $\rho_l = 1095\,\text{kg}\,\text{m}^{-3}$. The constants R and M_w are the molar gas constant and molar mass of water used by the Kelvin conversion. Since $\ln(\text{RH}) < 0$, the converted liquid pressure is negative in the same suction convention used by the active open-niche curve. This is consistent with $p_c = -p_l$ in the Richards model [22, pp. 448–452] [30, Sec. 3, Eqs. 1–2].

Uncertainty and reliability. The active OGS curve has 845 rows and spans approximately −53.24 to −5.23 MPa. The RH boundary check compares 2280 rows inside the active curve time range and finds large differences between RH-derived pressure candidates and the active curve. Therefore $p_E(t)$ should remain an existing OGS input, not a verified RH-derived forcing, until the source sensors, filtering, smoothing, time origin, interpolation, and extension policy are reconstructed.

Current computational use.

With the available source data now, RH/suction should be used as boundary-forcing provenance and uncertainty information for the open-niche pressure curve, not as an independent interior residual. The current best transfer is to convert screened RH records to a Kelvin pressure candidate with the stated constants, compare that candidate to the active XML curve $p_E(t)$, and represent the mismatch as a boundary-condition gate or scenario envelope. This is grounded in the active curve values, RH workbook check, and the CD-A water-content/RH formulation where negative liquid pressure represents suction through the chosen pressure convention [22, pp. 448–452] [30, Sec. 3, Eqs. 1–2] [5, RH source workbooks used for the generated boundary checks].

In the inverse problem, uncertain $p_E(t)$ is dangerous because permeability, retention, and boundary suction all control similar drying/wetting transients. If the boundary curve is treated as exact while its provenance is unresolved, the fit can falsely compensate by changing $\boldsymbol{K}(\boldsymbol{x})$ or retention parameters.

Grounding questions.

Table 2.6: Grounding questions for RH/suction and pressure-boundary forcing.

Question. Is RH an OGS input or output comparison?	
Policy	Primarily an input-side boundary source or check for $p_E(t)$, not an interior OGS output residual.
Question. How is RH converted to model pressure?	
Ready	By the Kelvin equation with RH as a fraction, a stated temperature, density, gas constant, molar mass convention, and $p_c = -p_l$ sign convention.
Question. Which sensors define the open-niche boundary?	
Gate	RH5/RH6 are the cleanest source candidates, but final use needs explicit sensor selection, filtering, high-RH/low-outlier handling, and time alignment.
Question. Why can $p_E(t)$ be negative?	
Ready	In this Richards convention it is liquid pressure relative to the gas/atmospheric reference; negative p_l corresponds to positive suction or capillary pressure.
Question. What would fully ground the pressure boundary?	
Gate	Source table, sensor selection, smoothing/filtering, interpolation and extrapolation rules, calendar time origin, sign/unit convention, and a covariance or scenario envelope for boundary uncertainty.

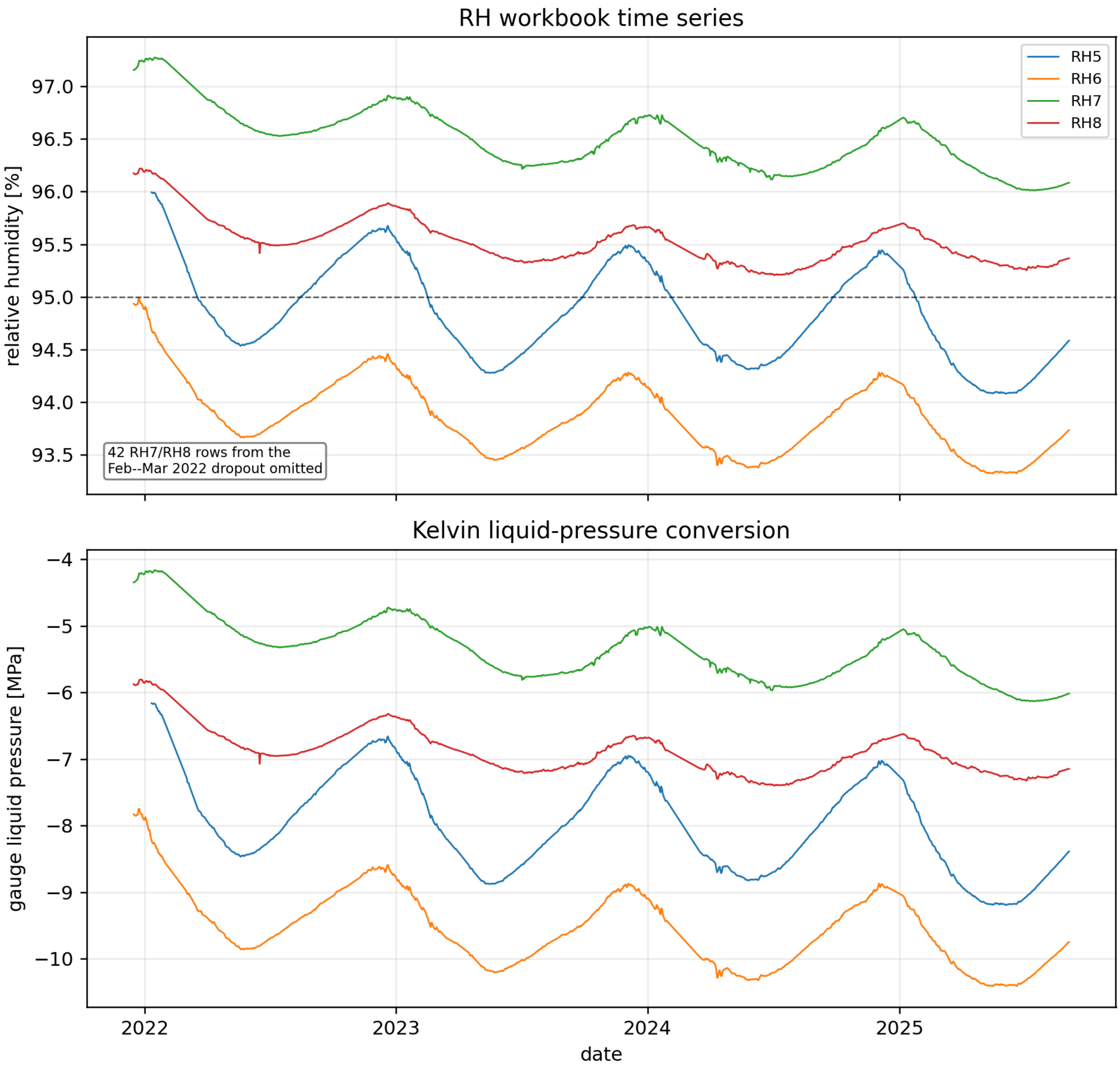

Figure 2.12: RH and Kelvin-converted pressure visualisation. The upper panel shows RH5–RH8 histories after omitting the flagged RH7/RH8 dropout interval; the lower panel shows the corresponding gauge liquid-pressure conversion in MPa.

2.7 Direct Pressure Measurements

Meaning. The direct-pressure stream is the mini-piezometer stream. In the CD-A measurement inventory, mini-piezometers measure pore pressure with unit Pa [29, Table 1]. The VEIS paper confirms that CD-A has borehole sensors for pressure measured by minipiezometers and that sensor databases store sensor names, date/time stamps, and measured values [10, Secs. 2.4 and 3.2]. The 2026 email-provided technical discussion identifies BCD-A28 to BCD-A31 as the current mini-piezometer set and shows their pressure trends in kPa [25, PDF p. 32].

Position in the domain. Figure 2.13 includes the source-slide location inset for A28–A31. That inset is enough to identify the instrument family and open/closed-twin context, but it is not yet an accepted OGS point set. A usable pressure residual still needs individual sensor coordinates or trace supports, sensor elevations/depths, and a rule for projecting each 3D sensor support to the present 2D slice.

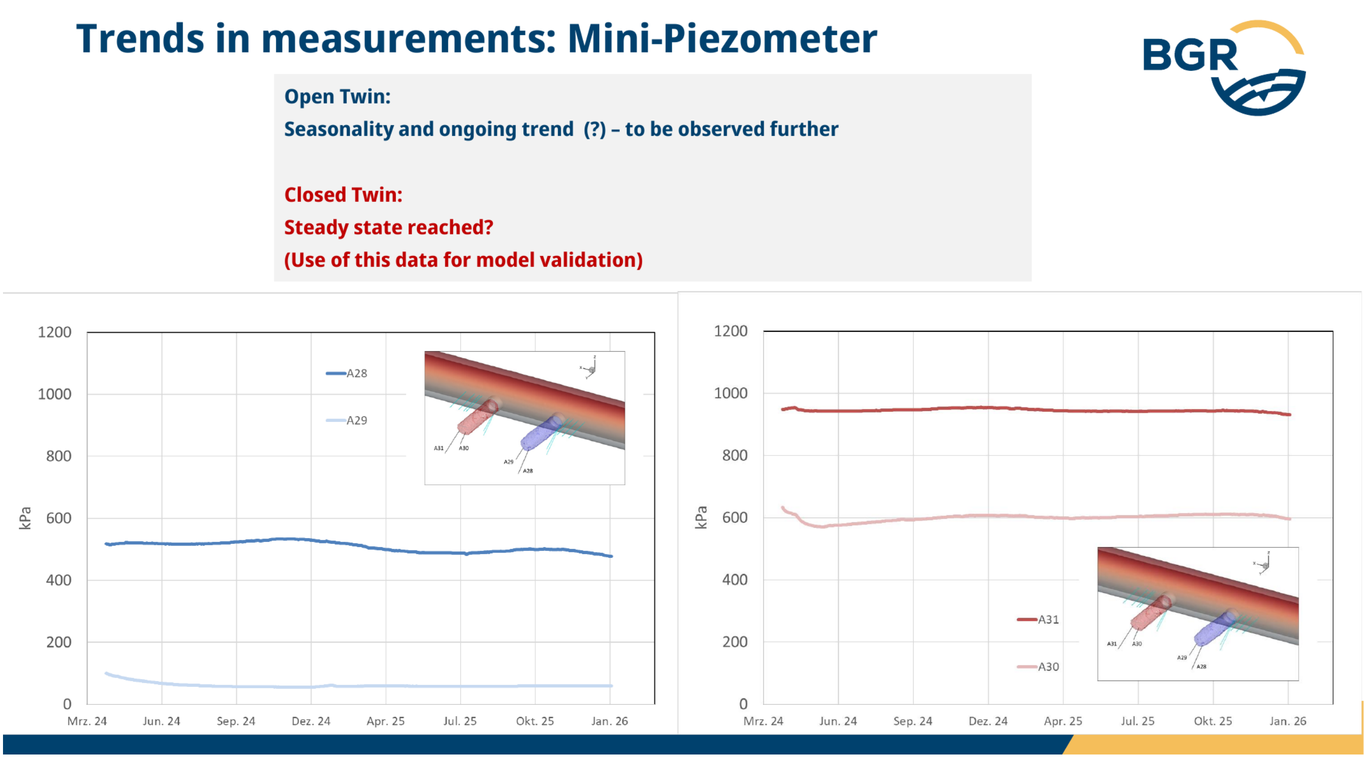

Figure 2.13: Mini-piezometer trend evidence from the 2026 technical discussion slides. The plotted values are in kPa and cover approximately March 2024 to January 2026. The open twin shows seasonality and an ongoing trend; the closed twin is presented as a possible steady-state validation stream. This is a source-slide plot, not a machine-readable export.

Time of measurement. The source plot spans about March 2024 to January 2026. The May 2026 minutes state that BCD-A28 to BCD-A31 were working well, and list pore-pressure measurements as later comparison targets for the modelling workflow [6, PDF p. 4 and p. 8]. The sampled numeric export behind the plot is still absent from the current provided catalogue, so exact timestamps, sampling interval, maintenance gaps, and quality flags remain unavailable.

Measured value, units, and OGS tie. The source plot uses kPa. The public CD-A inventory names the quantity as pore pressure in Pa, but the provided export must still state whether the stored values are absolute pressure, gauge pressure, hydraulic head, or another Geoscope conversion. Once that reference convention is known, the OGS tie is direct: sample $p_l(\boldsymbol{x}, t)$ at the sensor support and convert pressure unit, elevation head, atmospheric reference, and sensor depth consistently with the model pressure convention.

Uncertainty and reliability. This stream is promising but not residual-ready. The required export must include timestamps, units, sensor ids, coordinates or support ids, zero/reference convention, calibration, temperature compensation, quality flags, maintenance/failure flags, and raw-file provenance. The 2D model error may be significant if a mini-piezometer samples a local 3D feature or lies outside the effective support of the modeled slice.

Current computational use.

With the available source data now, direct pressure measurements are a blocked but high-value future residual. The source-slide trends prove that the stream exists and has physically meaningful evolution, but the paper does not yet have the machine-readable export needed for fitting. Once that export is available, the operator should be a direct sample or support average of OGS $p_l(\boldsymbol{x}, t)$, or a hydraulic-head comparison after elevation and reference-pressure conversion. This would directly test whether the far-field pressure, open-niche boundary curve, initial pressure field, and fitted permeability field are mutually consistent [29, Table 1] [10, Secs. 2.4 and 3.2] [25, PDF p. 32] [5, provided pressure/HM source files and generated trend figures].

The most important current modelling use is not to fit the slide image. It is to request the pressure export in a form that closes the pressure-reference ambiguity. Confusing absolute pressure, gauge pressure, hydraulic head, suction, or capillary pressure would shift the residual and can force incorrect changes in boundary pressure, initial pressure, or $\boldsymbol{K}(\boldsymbol{x})$.

Grounding questions.

Table 2.7: Grounding questions for direct pressure measurements.

Question. Do direct pore-pressure measurements exist for this model?	
Blocked	Yes as a documented mini-piezometer stream and source-slide trend; no residual-ready numeric export is present yet.
Question. How should pressure be linked to OGS?	
Gate	Compare to $p_l(\boldsymbol{x}, t)$ or hydraulic head only after units, elevation, absolute/gauge/reference convention, and sensor support are known.
Question. Can pressure data constrain initial and boundary conditions?	
Gate	Yes; it can separate boundary-condition error from permeability-field error and test the post-excavation initial pressure state.
Question. What 2D-model caveat is needed?	
Gate	Each sensor may sample a 3D hydraulic path or feature outside the modeled slice; the export must support a defensible 3D-to-2D comparison rule.
Question. What would fully ground pressure residuals?	
Blocked	Sensor coordinates, screened interval/depth/elevation, timestamps, raw units, pressure reference, calibration/maintenance flags, uncertainty, and raw-file provenance.

2.8 Faults, Fractures, and Crack Geometry

Meaning. This section is about large faults, fault zones, and mapped fracture geometry, not about crack-meter time series. The public CD-A characterization paper reports fault planes parallel to bedding as well as non-bedding-parallel fault planes, and summarizes the structural geology near the twin niches in its Fig. 4 [29, Sec. 3 and Fig. 4]. For the OGS workflow, these features are not observations of pressure or displacement by themselves. They are structural information that can affect permeability, porosity, EDZ extent, anisotropy, or prior covariance.

Position in the domain. The provided Tecplot layout contains 22 `Kluft` fracture/crack-geometry zones. Those zones document source-frame structural supports, but they do not yet define which 3D features intersect the current 2D model slice, what effective width should be used, or whether the feature should become a hard material input, a prior, a mask, or a qualitative caveat. The email-provided HERMES input note also states that geological characterization of fractured zones is available for CD-A [3, PDF p. 5].

Time of measurement. The geometry is structural, not a complete time series. A feature can still be used as time-dependent modelling information only if there is measured aperture, displacement, hydraulic effect, or dated structural evolution. Without that, it should be treated as static structural context or as a scenario/prior field.

Measured value, units, and OGS tie. For large faults and fracture zones, the model-facing quantities are 3D coordinates, plane or zone geometry, width/support, aperture or hydraulic effect, and a projection policy. The OGS tie is indirect: a fault may modify $\boldsymbol{K}(\boldsymbol{x})$, $n(\boldsymbol{x})$, EDZ classification, permeability-tensor orientation, or the prior covariance of those fields. It is not a residual against an OGS output unless a measured quantity, unit, time, and observation operator are defined.

Uncertainty and reliability. The main risk is overfitting a two-dimensional permeability field to a three-dimensional unresolved structure. A fault or crack should become a hard input only after the projection policy is explicit: which 3D feature intersects the 2D slice, what width/support is assigned, whether it changes $\boldsymbol{K}$, n, EDZ classification, or only the prior covariance, and which uncertainty represents projection error.

Current computational use.

With the available source data now, large faults and fracture geometry should enter as structural priors, masks, or scenario definitions rather than residuals. The provided `.dat`/Tecplot geometry gives source-frame structural zones, and the public CD-A characterization supports the relevance of faults and fractured zones, but geometry alone does not define hydraulic aperture, permeability contrast, or a time-dependent material change. The best current use is to construct alternative 2D projection scenarios: no fault effect, a projected EDZ/fault mask, a local permeability-basis support, or a covariance modifier for $\boldsymbol{K}$ and possibly n or tensor orientation [29, Sec. 3 and Fig. 4] [3, PDF p. 5] [5, geometry source file `VisualisationCDA.dat`].

For inversion, the key discipline is not to turn fault geometry into a fake measurement. A fault can guide where material fields are allowed to vary. It becomes a residual only if a numeric aperture, displacement, pressure response, or other measured value is supplied with units, time, support, and an observation operator.

Grounding questions.

Table 2.8: Grounding questions for large faults and fracture geometry.

Question. What does the large-fault source measure?	
Diagnostic	Currently it is structural geometry; no accepted residual-ready aperture, displacement, or hydraulic-effect table is present.
Question. How should faults tie to OGS now?	
Policy	As projected structural priors, EDZ/fault masks, permeability-basis supports, tensor-orientation context, or scenario fields.
Question. Can a fault modify permeability or porosity?	
Gate	Yes as a scenario or prior, but hard modification needs independent evidence from permeability, ERT, pressure, NMR/Taupe trends, or crack/HM data.
Question. How should 3D geometry be reduced to the 2D slice?	
Gate	By an explicit intersection, influence-band, or nearest-distance projection with effective width and uncertainty; otherwise it remains a caveat.
Question. What would fully ground quantitative fault use?	
Gate	Exact source file, coordinates and units, feature identities, 3D-to-2D projection, width/support, target fields ($\boldsymbol{K}, n,$ EDZ), time-dependence assumption, and independent hydraulic or mechanical evidence.

2.9 Geoscope, Crackometer, and Other HM Monitoring

Meaning. The other-HM package covers measurements that can constrain mechanics, pressure, swelling, permeability plausibility, or boundary disturbances: extensometers, crackmeters, laser scans, miniprisma geodesy, convergence points, precision levelling, evapometer or climate context, and Geoscope auxiliary streams. The CD-A inventory identifies extensometers and convergence as displacement measurements, laser scans as niche-profile displacement measurements in micrometres, and mini-piezometers as pore pressure measurements [29, Table 1]. The email-provided HERMES input note frames the same bundle as long-term deformation, humidity, water-content, pore-water-pressure, crack-width, piezometer, extensometer, ERT, TDR, and laser-scan measurements [3, PDF p. 5].

Position in the domain. The source layout contains support geometry for several HM streams, but the generic support plots are not useful paper figures because they do not show measured values, timestamps, units, or OGS-ready support. Table 2.9 therefore records the useful status directly. Before fitting, each stream needs a mapping from source frame to the current 2D model frame and, where relevant, a 3D-to-2D comparison rule.

Table 2.9: Other-HM measurement status in the current source catalogue.

Stream	What it measures	Current source evidence	OGS tie and status
Extensometer	Displacement or strain along instrument geometry.	Geoscope/minutes evidence; BCD-A9/B10 closed-niche failure since September 2025.	$\boldsymbol{u}$ or strain diagnostic; hard residual blocked until time series, geometry, sign, and quality flags are supplied.
Crackmeter	Crack aperture or local relative displacement.	Source trend plot and minutes: one closed-twin trend and open-twin seasonal variation.	$\boldsymbol{u}$, strain, or crack-aperture diagnostic; residual blocked until positions, zero date, sign convention, units, and uncertainty are supplied.
Laser scans	Registered niche-surface displacement or scan-difference fields.	Minutes state a 2026-04-20 statistical laser-scan update was sent; the provided catalogue has only surface geometry.	Surface-displacement diagnostic; blocked until registered products, masks, reference epochs, and registration uncertainty are supplied.
Levelling	Vertical displacement at survey points.	Twelve slide-summary values from the 2022-08-29/30 reference to the 2026-03 campaign.	Vertical $\boldsymbol{u}$ validation target; diagnostic only until full survey table, coordinates, reference frame, and covariance are supplied.
Auxiliary Geoscope / evapometer	RH, temperature, door/opening times, suction, evaporation or climate context.	Minutes and support geometry; no residual-ready auxiliary export here.	Boundary-condition provenance and disturbance filtering; not an interior OGS residual by default.

Time of measurement. The provided package contains qualitative 2026 statements, a source crackmeter trend plot, a source mini-piezometer trend plot, and one numeric levelling slide summary. The May 2026 minutes state that the 2026 Geoscope update covered RH, temperature, door opening times, extensometer, mini-piezometer, crackmeter, suction, and laser scans, and that horizontal/vertical extensometers BCD-A9/B10 in the closed niche failed after September 2025 [6, PDF p. 4]. The levelling slides report height differences from the first survey on 2022-08-29/30 to the 2026-03 campaign [7, PDF pp. 2, 5].

Measured value, units, and OGS tie. Potential model ties are straightforward once data exist. Extensometers constrain axial displacement or strain from $\boldsymbol{u}$. Crackmeters constrain crack aperture or relative displacement. Laser scans constrain surface displacement or scan-difference fields. Miniprisma and convergence points constrain point or pair displacement. Levelling constrains vertical displacement. Evapometer or climate proxies should enter only through boundary forcing, not as interior residuals.

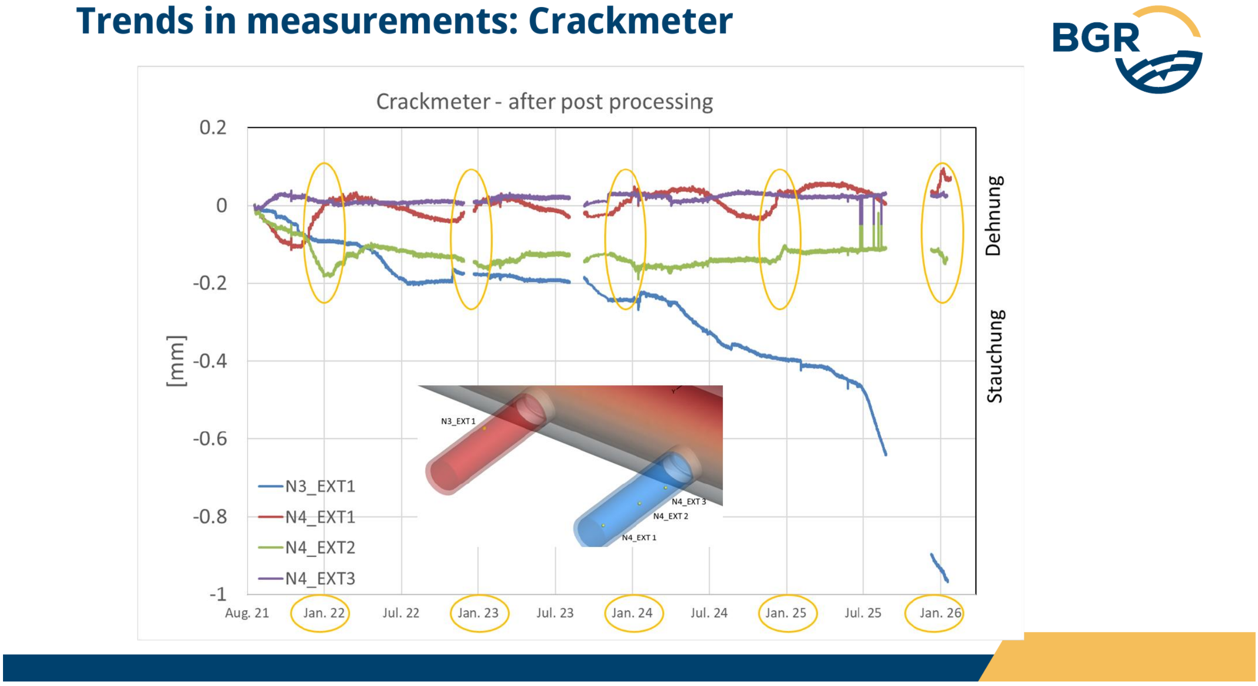

Figure 2.14: Crackmeter trend evidence from the 2026 technical discussion slides. The plotted post-processed deformation is in millimetres and spans roughly August 2021 to January 2026. The slide shows seasonal markers and an instrument inset. It is useful trend evidence, but not a residual-ready export because the paper still lacks positions, zero/reference convention, sign convention, uncertainty, and quality flags.

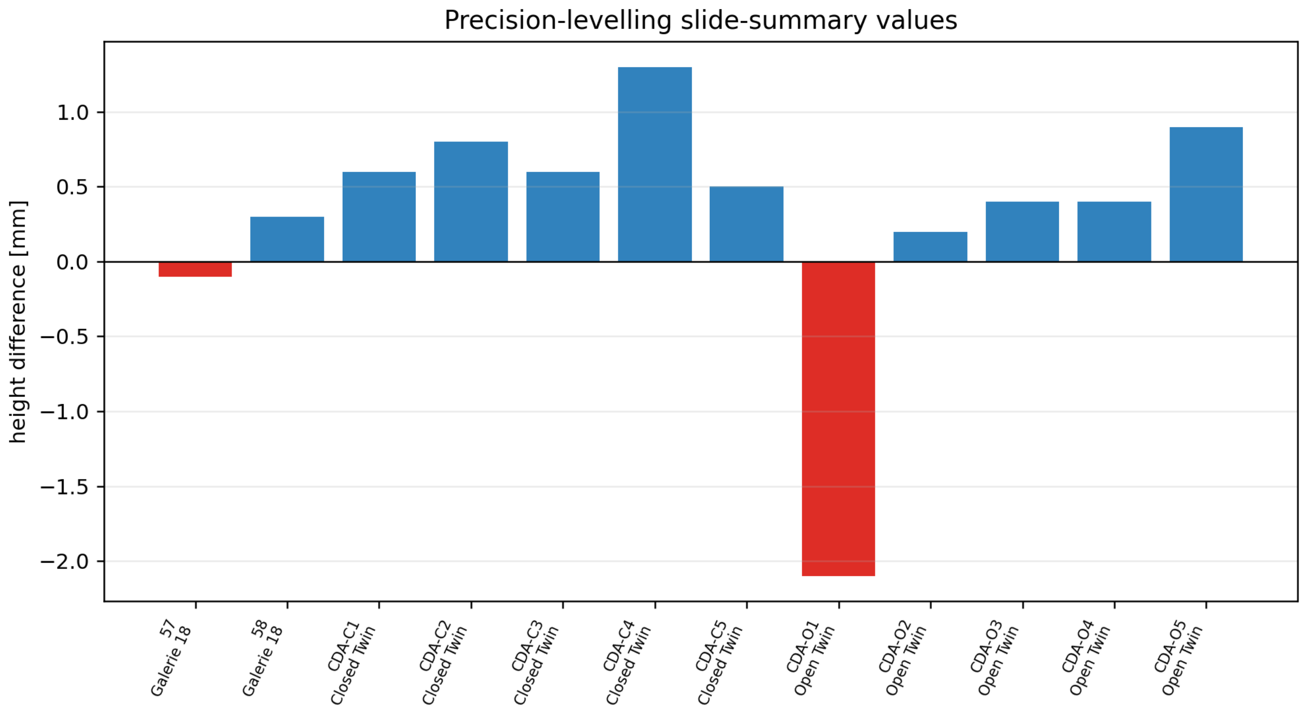

Figure 2.15: Precision-levelling values extracted from the slide summary. They give height differences between the 2022 initial survey and the 2026-03 campaign, but not a full survey table or covariance.

Uncertainty and reliability. The HM streams remain mostly blocked for hard residuals. The source scan found layout geometry, laser surface geometry, levelling slide rows, and report/minute evidence, but no residual-ready Geoscope time series, laser statistical product, or full levelling survey table [5, provided HM source files and generated trend figures]. A hard residual requires timestamps or survey epochs, values, units, support geometry, reference-zero conventions, sign conventions, calibration, quality flags, failure flags, and uncertainty or covariance. The levelling slide gives a detectable displacement of 0.1–0.2 mm at 95%

confidence, largest settlement at CDA-O1 (−2.1 mm), and largest uplift at CDA-C4 (+1.3 mm), but it still lacks the full survey table and reference-frame metadata needed for weighting [7, PDF pp. 5–6].

Current computational use.

With the available source data now, niche-surface cracks and other HM streams should mostly be validation gates, qualitative priors, or future residual packages. The crackmeter source plot can test whether a model candidate gives plausible opening or closure trends only after the measured component, zero date, sign convention, instrument axis, and support are known. Extensometers, convergence, miniprisma, levelling, and laser scans have clear OGS displacement ties, but the current workflow keeps mechanical parameters and prestress fixed. Therefore these streams should first reject mechanically implausible hydraulic candidates, support EDZ/fracture scenarios, or define quality/disturbance flags; they should not yet drive a coupled HM calibration [29, Table 1] [26, abstract and Sec. 2] [6, PDF p. 4 and p. 8] [5, provided HM source files and generated trend figures].

If numeric exports are supplied, the operators are standard but stream-specific: extensometers use projected displacement or strain along an instrument axis; convergence uses point-pair displacement; crackmeters use relative displacement or crack opening; levelling uses vertical displacement; laser scans use registered surface differences with masks and registration covariance. Auxiliary climate, door, temperature, evapometer, or Geoscope context should enter as boundary provenance, quality flags, or disturbance filters rather than interior residuals.

Grounding questions.

Table 2.10: Grounding questions for Geoscope, crackometer, and other HM monitoring.

Question. What does each HM stream measure?	
Gate	Extensometers measure axial deformation, crackmeters measure crack opening or relative displacement, levelling measures vertical displacement, laser scans measure surface differences, and auxiliary streams describe boundary or quality context; each source export must confirm the exact quantity.
Question. How should crackmeters tie to OGS?	
Gate	Compare to relative displacement $(\boldsymbol{u}_a - \boldsymbol{u}_b) \cdot \boldsymbol{n}$, where $\boldsymbol{n}$ is the local crack-normal or instrument-axis direction, near-crack strain, or a crack-aperture diagnostic only after the component and sign convention are known.
Question. Can visible cracks justify permeability changes?	
Policy	Yes as an EDZ/fracture prior or scenario, but not as a weighted residual unless a measured aperture or displacement operator is supplied.
Question. How should levelling, extensometers, and laser scans be used now?	
Diagnostic	As diagnostic HM validation or rejection gates for candidate permeability and boundary scenarios, because mechanical parameter release is not currently grounded.
Question. Which other measurements can enter the OGS workflow?	
Policy	Any exported pressure, displacement, strain, crack-width, laser, climate, temperature, evapometer, geology, or support-geometry stream can enter as an input, residual, prior, mask, quality filter, or caveat once its support, time, units, and uncertainty are explicit.
Question. What would fully ground HM residuals?	
Blocked	Machine-readable values with timestamps or survey epochs, coordinates/supports, units, reference-zero and sign conventions, instrument axes, failure/quality flags, uncertainty or covariance, and a 3D-to-2D comparison policy.

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